MouRa - Agri AI



STA 201 STATISTICAL METHODS (1+1)

LECTURE NOTES
B.Sc (Hons.) Agriculture

COURSE TEACHER

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STA 201 Statistical Methods (1+1)

Theory

Unit I: Descriptive Statistics

Basic concepts – statistics – variable – types and sources of data – classification and tabulation of data. Diagrammatic and graphical representation of data – simple, multiple, component and percentage bar diagrams, pie diagram – frequency polygon, frequency curve and histogram. Construction of frequency distribution tables.

Measures of central tendency: arithmetic mean, geometric mean, harmonic mean, median and mode – merits and demerits. Measures of dispersion: range, quartile deviation, mean deviation, standard deviation, and coefficient of variation – skewness and kurtosis – merits and demerits.

Unit II: Probability Distributions and Sampling Theory

Probability – basic concepts – additive and multiplicative laws (without proof). Probability distributions – Discrete distributions: Binomial and Poisson. Continuous distribution: Normal distribution – definitions and properties.

Sampling theory – population – sample – parameter and statistic – sampling distribution – sampling vs complete enumeration – Types of sampling – simple random sampling – selection of simple random sample using random number tables.

Unit III: Testing of hypotheses

Null and alternative hypothesis – types of errors – critical region and level of significance – degrees of freedom. Large sample test – single proportion and difference between two proportions – single mean and difference between two means.

Small sample tests – F-test – t-test for testing the significance of single mean – independent t test and paired t test – chi square test for goodness of fit – chi square test for testing the association of attributes by m x n contingency table – 2 x 2 contingency table – Yates' correction for continuity.

Unit IV: Correlation and Regression

Correlation – Scatter diagram – Karl Pearson's correlation coefficient definition – computation – types of correlation and properties. Regression – simple linear regression – fitting of simple linear regression equation – properties of regression coefficient.

Unit V: Analysis of Variance and Experimental Designs

Analysis of Variance (ANOVA) – assumptions – one way and two way classifications. Basic principles of experimental designs – Completely Randomized Design (CRD) – Randomized Block Design (RBD) – Latin Square Design (LSD) – lay out, analysis, merits and demerits of the above mentioned designs.

Practical

Formation of frequency distribution tables – Diagrammatic and graphical representation. Computation of different measures of central tendency and computation of various measures of dispersion for raw and grouped data – calculation of coefficient of variation (CV) – measures of skewness and kurtosis. Simple problems in Binomial distribution, Poisson and Normal distribution – Selection of simple random sampling. Large sample test

for single proportion and difference between two proportions and Large sample test for single mean and difference between two means. t-test for single mean – t-test for testing the significance of two means for independent and paired samples – chi square test for goodness of fit and test for independence of two attributes in a contingency table – Yates correction for continuity

– calculation of the correlation coefficient – fitting of simple linear regression equation – One way and two way ANOVA – completely randomized design (CRD) – randomized block design (RBD) – Latin square design (LSD).

Theory Lecture Schedule

1. Basic concepts – statistics – variable – types and sources of data – classification and tabulation of data. Diagrammatic and graphical representation of data – simple, multiple, component and percentage bar diagrams, pie diagram – frequency polygon, frequency curve and histogram. Construction of frequency distribution tables.
2. Measures of Central Tendency – meaning – limitations – properties – mean, median mode geometric mean and harmonic mean for ungrouped and grouped data.
3. Measures of Dispersion – meaning – limitations – properties – range and mean deviation, Quartile deviation, standard deviation, variance and coefficient of variation for ungrouped and grouped data. Skewness and kurtosis – types – uses.
4. Probability – basic concepts – axioms – mathematical and statistical probabilities – additive and multiplicative laws (without proof). Theoretical discrete distributions – Binomial and Poisson distribution and its applications.
5. Theoretical continuous distribution – Normal distribution and its properties and importance – standard normal distribution.
6. Sampling theory – population – sample – sampling vs complete enumeration – parameter and statistic – need for sampling – sampling distribution – standard error.
7. Sampling methods – probability sampling method – simple random sampling – Selection using random number tables and lottery method.
8. Tests of significance – basic concepts – null and alternative hypotheses – critical region – level of significance – degrees of freedom.
9. **Mid Semester Examination**
10. Large sample test – single proportion and difference between two proportions – single mean and difference between two means
11. Small sample tests – F-test – t-test for testing the significance of single mean independent t test and paired t test
12. Chi square test for goodness of fit – chi square test for testing the association of m x n contingency table

    – 2 x 2 contingency table – Yates' correction for continuity
13. Correlation – Scatter diagram – Karl Pearson's correlation coefficient definition – computation – types of correlation and properties.
14. Regression – simple linear regression – fitting of simple linear regression equation – properties of regression coefficient.

15. Analysis of Variance (ANOVA) – assumptions – one way and two way classifications. Basic principles of experimental designs – randomization, replication and local control.
16. Completely Randomized Design (CRD) – Randomized Block Design (RBD).
17. Latin Square Design (LSD).

Practical schedule

1. Construction of frequency distribution tables.
2. Diagrammatic representation – simple, multiple, component and percentage bar diagrams, pie diagram. Graphical representation – frequency polygon, frequency curve and histogram.
3. Computation of arithmetic mean, geometric mean, harmonic mean, median and mode for ungrouped and grouped data.
4. Computation of range, standard deviation, variance, coefficient of variation for ungrouped and grouped data. Computation skewness and kurtosis for ungrouped and grouped data.
5. Simple problems in Binomial distribution and Poisson distribution.
6. Simple problems in Normal distribution.
7. Selection of simple random sample using simple random sampling method.
8. Large sample test – test for single proportion and difference between two proportions.
9. Large sample test – test for single mean and difference between two means.
10. Small samples test – t-test for single mean – independent t test for difference between two sample means (equal variances only) – Paired t-test.
11. Chi square test for goodness of fit – Chi square test for testing the association of attributes.
12. Computation of Karl Pearson's correlation coefficient.
13. Fitting of simple linear regression equation y on x.
14. One way ANOVA – analysis of experimental data using Completely Randomised Design (CRD) (for equal replications only).
15. Two way ANOVA – analysis of experimental data using Randomised Block Design (RBD).
16. Analysis of experimental data using Latin Square Design (LSD).
17. **Final Practical Examination**

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STA 201-STATISTICAL METHODS(1+1)

LECTURE SCHEDULE-1

The word 'statistics' comes from the Italian word 'statista' (meaning "statesman") or the German word 'statistik' each of which means a political state. It was first used by Professor Gottfried Achenwall (1719-1772), a professor in Marlborough in 1749 to refer to the subject-matter as a whole. Achenwall defined statistics as "the political science of the several countries" to help the government of rule, plan policies in peace and war (or) to collect taxes.

It has several meanings, one meaning is "numerical/quantitative figures" Examples are number of crops in a farm, number of students in a class and number of unemployed persons in a district etc. Another meaning of statistics is "Body of scientific principles and techniques". Based on that several definitions are given by different authors as follows.

Definitions of Statistics given by different authors

1.	**HORACE SECRIST**	**:**	'By statistics we mean aggregates of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated according to reasonable standards of accuracy, collected in a systematic manner for a pre-determined purpose and placed in relation to each other!.'
2.	**KING**	**:**	'The science of statistics is the method of judging collective natural or social phenomenon from the results obtained from the analysis of enumeration or collection of the estimates'
3.	**BUDDINGION**	**:**	Statistics is a science of **estimates and probabilities**.
4.	**A.L. BOWELY**	**:**	Statistics may be called as **a science of counting**.
5.	**COWDEN and CROXTEN**	**:**	Statistics defined as **collection, analysis and interpretation of data.**

In General

It can be defined as the body of concepts, principles and methods dealing with collection, summarisation, analysis and interpretation data.

Data

Collection of data points is known as Data.

Primary data

The data collected primarily for a particular study, called primary data.

Eg: census data.

Secondary Data

The data collected for some other purpose and it is used for another studies, called secondary data or second hand data.

Methods of collecting primary data

i) Interview method

ii) Mailed questionnaire method.

Interview method

In this method, the investigator from house to house interviews the individuals personally. He asks the questions one by one and fills up the schedule on the basis of the information supplied by the individuals.

Mailed questionnaire method

In this method, the questionnaire is mailed to the individuals who are required to fill them up and return it duly completed.

Qualitative data

If the observations of an experiment are in qualitative nature called qualitative data.

Eg.: Colour of a flower, taste of a fruit, shape of the seed etc.,

Quantitative data

If the observations of an experiment are in quantitative nature, called quantitative data.

Eg.: Yield of a crop, number of unemployed persons in our nation, rain fall over a period of time etc.,

Inclusive class

In a class, if the upper limit of a class is not equal to the lower limit of the immediate succeeding class then it is called inclusive class

Eg.: 10 – 20

21 – 30

31 – 40

41 – 50

Exclusive class

In a class, if the upper limit of class is equal to the lower limit of immediate succeeding class is called exclusive class.

Eg.: 10 – 20

20 – 30

30 – 40

40 – 50

Frequency distribution

Summarization of data starts with classification of the data. Arranging of data according to some common features is known as classification. Broadly classification is done according to the type of data ie., qualitative or quantitative data. In case of qualitative data, we count items with specified character and put them in appropriate categories. If the data are measurable (ie quantitative), they are classified according to classes, Sorting of data into categories or classes will lead to formation of frequency distribution .It is a quantitative or qualitative classification of data.

Construction of frequency distribution for quantitative data

Step 1 : The number of classes will be decided by Yule's formula

$$\text{Number of Classes} = 2.5 \text{ X } (n)^{1/4}$$

Step 2: The class interval will be decided by the relation

$$\text{Class interval} = \frac{\text{(Maximum value in the data set) - (Minimum value in the data set)}}{\text{Number of classes}}$$

Step 3: Construct a rough frequency distribution by using exclusive classes with tally bars.

Step 4: Construct a fair final frequency table by using exclusive classes.

Step 5: A self-explanatory title should be given at the top or bottom of the frequency table.

Cumulative frequency curve: (or Ogive)

The cumulative frequency of a class is the total frequency up to and including that class. The table of cumulative frequency is called cumulative frequency distribution. There are two type of cumulative frequency distribution. The cumulative frequency of all values greater than or equal to the lower limit of each class is called more than cumulative frequency distribution. The cumulative frequency of all values less than to the upper limit of each class is called less than cumulative frequency distribution.

In drawing frequency curve, if the cumulative frequencies are taken then the curve is said to be Cumulative frequency curve (or) Ogive. If it is greater than type then the curve is said be greater than Ogive. If it is less than type then the curve is said to be less than Ogive.

Variable

A quantitative or qualitative characteristic that varies from observation to observation in the same group in called a variable. Generally quantitative variables are called variables.

Eg. Yield of a crop, height of the students etc.,

Attribute

The qualitative variables are called attributes.

Eg. : Crop varieties, Soil types, Shape of seeds, Colour of flowers etc.,

Population

The aggregate or totality of all possible objects possessing the specified characteristic is called a population.

Eg: If some information is required about an average fourth year B.Sc.(Ag.) student in Agricultural college & Research Institute Madurai, then the collection of fourth year B.Sc. (Ag) students in AC & RI, Madurai will be the population.

Sample

It is defined as smaller part of population.

Eg. : If collection of fourth year students in AC & RI, Madurai is the population, then the students in 'A' batch of fourth year will be the sample.

Characteristics of statistics	**:**	1. They must be numerical statements. 2. They must be aggregates of facts. 3. They must be related to each other. 4. They must have certain objectives behind them. 5. They must be affected to a marked extent by multiplicity of causes. 6. They are estimated with a reasonable standard of accuracy.
Uses of statistics	**:**	1. Simplifies complexity of the data. 2. Measures the results like averages etc. 3. Studies the relationship. 4. Useful for estimations. 5. Useful for surveys. 6. Useful in Agricultural and biological sciences medical sciences research.
Limitations of statistics	**:**	1. Statistics does not study qualitative data. i.e. It studies only numerical figures. 2. Statistics does not study individuals. i.e. It studies only aggregates.

3. Statistics is liable to be misused.
4. Statistical results some times lead to wrong conclusions unless they are properly used.
5. Statistical laws are not exact.

Solved Problems

Steps involved for construction of frequency table:

Step 1 **:** Find out the maximum and minimum values for the given data and calculate range(R).

R = maximum value – minimum value.

Step 2 **:** Decide the number of class. For this, Yule's formula is used.

No. of class (N) = $2.5 \times n^{1/4}$, n – total number of observation.

Step 3 **:** Find out the Class Interval (C.I)

C. I = R / N , R- Range; N-Number of classes.

Step 4 **:** Construct a rough frequency table with exclusive or inclusive classes using tally bars or tally marks.

Tally marks are used to transfer the data to frequency table.

Step 5 **:** Construct a fair frequency table with exclusive and inclusive classes and give suitable title at the top or bottom of the table.

DIAGRAMS & GRAPHS

Introduction

The visual aids, like diagrams and graphs, are helpful in understanding the facts of numerical data easily. The salient features of the data can be known at a glance by using these visual aids. Comparison of data is also made easier by such diagrams and graphs.

Advantages:

i) They are more attractive

ii) They give complete analysis of the given data at a glance. ie., they are more eye-catching.

iii) They are more impressive.

iv) They are useful in the comparison of two or more data sets.

Disadvantages:

i) The charts/graphs con not show as many facts as can be shown in a table
ii) Exact values con not be given in a chart/graphs
iii) Charts require a certain amount of time to construct

Differences between diagrams and graphs

There is no clear cut definition about the differences between diagrams and graphs. But some points may be remembered while drawing diagrams and graphs.

i) Generally speaking graphs are drawn on graph paper while diagrams are on plain sheet.
ii) A graphs represents mathematical relationship of two variables where as a diagram does not.
iii) Diagrams are not helpful in the analysis of the data but graphs are.
iv) To represent a frequency distribution, graphical representation are preferable than diagrams.

Important diagrams

i) Simple bar diagram
ii) Multiple bar diagram
iii) Component bar diagram
iv) Percentage bar diagram
v) Pie-chart

Important graphs

i) Histogram
ii) Frequency curve
iii) Frequency polygon
iv) Ogive

Simple bar diagram

It consists of vertical bars of equal width. The heights of the bars are proportional to the volume or magnitude of the attribute. All bars stand on the same

base line. The bars are separated from each other by equal intervals. The bars may be coloured are marked.

Multiple bar diagram

It is simply the extension of simple bar diagram. For each attribute two or more bars to be placed side by side. Each bar within a attribute will be marked or coloured differently in order to distinguish them. A foot note has to be given explaining the markings or colourings.

Component bar diagram

It is also called sub-divided bar diagram. Here the bars are placed one on top of the other. These diagrams are used to represent various parts of a whole. Each component will be marked or coloured differently. A foot note has to be given explaining the markings or colourings.

Percentage bar diagram

It is a type of component bar diagram. Here the frequencies are converted into percentage. The total frequency of each attributed is taken as 100 percentage. Therefore the heights of rectangular bars are same.

Pie chart

It is a circular diagram. It may be used in place of bar diagrams. Here the frequencies are converted into angles by taking 360 degrees as whole. For a convenient radius a circle is drawn and it will be divided into many sectors according to the converted angles. These sectors may be marked or coloured differently. A foot note has to be given explaining the markings or colourings.

Histogram

When the data are classified based on the class – intervals, it can be represented by a histogram. Histogram is just like a simple bar diagram with minor differences. There is no gap between the bars since the classes are continuous. The bars are drawn only in outline without colouring or making as in the case of simple bar diagram.

Frequency polygon

It is a graph of frequency distribution. There are two ways to construct a frequency polygon.

i) In a histogram if we join the mid points of the top of these rectangular bars we get a polygon. Such a polygon is called a frequency polygon.

ii) A frequency polygon can also be drawn directly without constructing the histogram. This is done by plotting the mid points with frequencies as co – ordinates on a graphs sheet. These points are joined by straight lines to get the frequency polygon.

Frequency curve

The procedure for drawing a frequency curve is same as for frequency polygon. But the points are joined by smooth or free – hand curve. The curve may not pass through all points as in case of frequency polygon.

LECTURE SCHEDULE-2

MEASURES OF CENTRAL TENDENCY

Measures of central tendency help to get a single value that describes the characteristics of the entire mass of data. Such a value is called the central value

There are five measures of average value of a variable. These measures are also known as **measures of central tendency** (or) **measures of location** (or) **averages** because these represent the **centre** of the distribution.

The **five measures** are

1. Arithmetic Mean (AM)
2. Geometric Mean (GM) } → Computed averages
3. Harmonic Mean (HM)
4. Median
5. Mode } → Positional averages

Advantages of averaging

i) It will reduce the complexity of the data

ii) It will become a useful tool for the comparisons of two or more data sets.

Characteristics of a good average

i) It should be well defined.

ii) It should be based on all observations.

iii) It should be suitable for further algebraic treatment.

iv) It should be easy to calculate.

v) It should be easy to understand.

vi) It should not be affected by sampling fluctuation

ARITHMETIC MEAN (AM)

It is a method of finding the average is to take the sum of all observations by the total number of observations. This is generally denoted by $\bar{x}$.

$$\text{Mean} = \bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n} \longrightarrow \textbf{Raw data (or) Ungrouped data}$$

Where n – is the number of observations

$$\text{Mean} = \bar{x} = \frac{\sum fx}{\sum f} \longrightarrow \textbf{Grouped data-Discrete case}$$

Where , n- total frequency = Σ f , f- frequency

Mean= $\bar{x} = A + \left(\frac{\sum fd}{\sum f} \right) \times c$ ⟶ **Grouped data-Continuous case**

Where $d = \frac{midx - A}{c}$, A-assumed mean, c-class interval length

Example 1 Calculate the mean for 2, 4, 6, 8, 10

$$\bar{x} = \frac{2+4+6+8+10}{5}$$

$$= \frac{30}{5} = 6$$

Merits, Demerits and uses of Arithmetic Mean

Merits of arithmetic mean

i) It is well defined

ii) It is based on all observations

iii) It is easy to calculate

iv) It is easy to understand

v) It is suitable for algebraic treatment

vi) It is least affected by fluctuations of sampling.

Demerits

i) It is affected by extreme values

ii) It cannot be determined by inspection

Uses of Arithmetic Mean

It is considered to be the best of all averages. Arithmetic mean is called the **ideal average.** It is used in social, economic and business problems. The expressions like 'Average cost of production', 'Average price' and 'Average yield per acre' etc have the idea of none but the arithmetic mean.

Properties of Mean

1. Sum of the deviations taken from the mean is zero
2. If each observation is increased by k then the mean of the new observations is the original mean increased by k
3. If each observation is decreased by k then the mean of the new observations is the original mean decreased by k

4. If each observation is multiplied by k $(k \neq 0)$ then the mean of the new observations is the original mean multiplied by k
5. If each observation is divided by k $(k \neq 0)$ then the mean of the new observations is the original mean divided by k

MEDIAN

It is the middle most value or the arithmetic mean of the two middle values of the given data after arranging the data in ascending descending order

Raw data (or) ungrouped data

Step -1: Arrange the data in ascending or descending order.

Step– 2: **For odd Numbers**,

$$\text{Median}=\left(\frac{n+1}{2}\right)^{th} \text{ observation}$$

where, n– total number of observation.

For even number,

$$\text{Median = Average of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+1\right)^{th} \text{ observations}$$

Grouped data: discrete case

$$Median=\left(\frac{N+1}{2}\right)^{th} term$$

$$\text{Where } N=\sum f$$

Grouped data: Continuous case

$$Median=l+\left(\frac{\frac{N}{2}-m}{f}\right)\times c$$

Where

l= Lower limit of median class

N = Total frequency

m = Cumulative frequency of pre-median class

f – Frequency of the median class

c- Width of the median class

Example 2

If the weights of sorghum ear heads are 45, 60, 48, 100, 65 gms. Calculate the median

Here n=5

In ascending orders: 45, 48, 60, 65, 100

Median= $\left(\frac{n+1}{2}\right)^{th}$ value

= $\left(\frac{5+1}{2}\right)^{th}$ value =60

Example 3

If the sorghum ear- heads are 5, 48, 60, 65, 65, 100 gms. Calculate the median**.**

Solution

Here n=6

Median = Average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2}+1\right)^{th}$ value

$$= \frac{\left(\frac{6}{2}\right)^{th} value + \left(\frac{6}{2}+1\right)^{th} value}{2}$$

$$= \frac{60+65}{2} = 62.5 \text{ gms}$$

Merits and Demerits and uses of Median

Merits of median

i) It is easy to calculate

ii) It is unaffected by extreme values

iii) It can be determined by inspection

iv) Its value generally lies in the distribution.

v) Median can be calculated even from qualitative phenomena like honesty, character etc.

Demerits

i) It is not well – defined

ii) It is not based on all observations

iii) It is affected by fluctuations of sampling

iv) It is not suitable for algebraic treatment.

Uses of Median

Median is especially useful in case when the items are not susceptible to measurement in definite units; example, in cases of intelligence, ability, efficiency etc. Median is useful in distributions where extreme classes are ill defined like "less than 10" or "more than 80" or "over 80" etc.

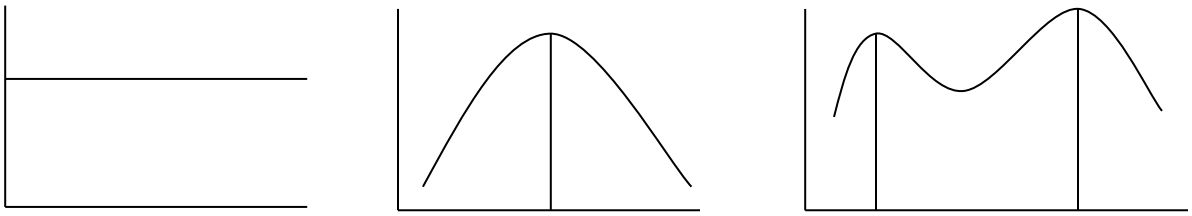
Properties of Median

i) The sum of the absolute values of deviation of the item from the median is minimum

ii) It is a positional average and is not influenced by the position of the items

MODE

Mode is defined as that item which occurs most frequently. For a frequency distribution, the mode is nothing but **the variate value** 'x' having maximum frequency. For a frequency distribution the mode may or may not exist. Even if exists it may be uni-modal, bi-modal (or) multi-modal.



No modal Uni-modal Bi- modal

Raw data (or) ungrouped data & Grouped data – Discrete case: Mode can often be found out by mere inspection in case of individual observations. The data have to be arranged in the form of an array so that the value which has the highest frequency can be known.

Case -1 : For raw data

Value which occur more frequently.

Case -2 : For discrete data

Value which has the highest frequency is called mode of the data.

Case – 3: For continuous data

$$Mode = l + \left(\frac{f - f_1}{2f - f_1 - f_2} \right) \times c$$

Merits and Demerits of Mode:

Merits

1. It is easy to understand as well as easy to calculate. In certain cases, it can be found out by inspection.
2. It is usually an actual value as it occurs most frequently in the series.
3. It is not affected by extreme values as in the average.
4. It is simple and precise.
5. It is the most representative average.
6. The value of mode can be determined by the graphic method.
7. Its value can be determined in an open end class-interval without ascertaining the class limits.

Demerits

1. It is not suitable for further mathematical treatment.
2. It may not give weight to extreme items.
3. It is not well defined
4. It is not based on all observations.
5. It is affected by fluctuations of sampling.
6. Mode is influenced by magnitude of the class-intervals.

Uses of Mode

The concept of mode is used by the people in their everyday life. For example, a manufacturer of banians, ready-made garments or shoes etc., is interested in the modal size and manufacture them in larger quantities. Mode helps the manufacturer in deciding the modals. It is useful in industry and business. Weather forecasts (Meteorological forecasting) are also based on mode. It is very useful to agriculturists, businessmen, etc. Mode is also used in socio-economic surveys. Mode is also mostly used in business and commerce.

Properties of Mode

i) It is not affected by the presence of extremely large or small items

ii) The mode is easily understood and easily ascertained in case of a discrete frequency distribution

Relationship between different averages:

i) In a symmetrical distribution, mean, median and mode will coincide.

i.e.) Mean = Median = Mode

ii) In an asymmetrical (skewed) distribution, these values will be different.

When positively skewed (skewed to the right) ⟶ Mean > Median > Mode

When negatively skewed (skewed to the left) ⟶ Mean < Median < Mode

iii) Mean – Mode = 3(Mean – Median)

iv) Mode = 3 Median – 2 Mean

v) If all the items in a distribution have the same value then,

Mean = GM = HM

vi) $A.M \geq G.M \geq H.M$

When to use different Averages

1. If data is qualitative only mode is used.

Example – Typical soil type in a locality or cropping pattern in a region.

2. Quantitative data we consider the nature of the frequency distribution.

Frequency is skewed – use median or mode

Symmetrical distribution - use mean or median or mode.

3. Raw data in which extreme values either small or large we use median or mode.

Geometric Mean:

It is the nth root of the product of n observations.

Case 1: Raw data

$$G.M = (x_1.x_2,...,x_n)^{\frac{1}{n}}$$

Take log on both sides,

$$\log G.M = \frac{1}{n}[\log x_1 + \log x_2 + ... + \log x_n]$$

$$\log G.M = \frac{1}{n}\sum \log x_i$$

$$G.M = anti\log\left(\frac{1}{n}\sum \log x_i\right)$$

Case – 2 : Discrete data

G.M = Antilog of (Σ f log x / N)

Where, N – total frequency, f – frequency.

Case – 3 : Continuous data

G.M = (Σ f log m / N)

Where, m – midpoint of each class, N – total frequency.

Example 4: If the weights of sorghum ear heads are 45, 60, 48,100, 65 gms. Find the Geometric mean for the following data

Weight of ear head x (g)	Log x
45	1.653
60	1.778
48	1.681
100	2
65	1.813
Total	8.925

Solution

Here n=5

$$G.M = anti\log\left(\frac{1}{n}\sum \log x_i\right)$$

$$= anti\log\left(\frac{8.925}{5}\right)$$

$$= anti\log(1.785) \quad = 60.95$$

Properties

1. All data must be positive zero data value will produce zero G.M
2. G.M is scale invariant toward the product of the factor
3. G.M<A.M
4. The product of the item remains unchanged if each item is replaced by the G.M
5. If the given values of a variable are equal then the G.M will be equal to their common value

HARMONIC MEAN

It is the reciprocal mean of the reciprocal observation.

For raw data

$$H.M = \frac{n}{\sum_{i=1}^{n}\left(\frac{1}{x_i}\right)}$$

Case – 2 : Discrete data

$$H.M = \frac{N}{\sum_{i=1}^{n}\left(\frac{f}{x_i}\right)}$$

Where, N– total frequency.

Case – 3 : Continuous data

$$H.M = \frac{N}{\sum_{i=1}^{n}\left(\frac{f}{m_i}\right)}$$

Where. m – midpoint, N – total frequency

Example: From the given data calculate H.M 5, 10, 17, 24, 30

x	$\frac{1}{x}$	Solution
5	0.2000	$H.M = \frac{5}{0.4338}$
10	0.1000	
17	0.0588	
24	0.0417	=11.526
30	0.0333	
Total	**0.4338**	

Properties

i) Harmonic mean is always smaller compared to G.M, A.M or quadratic mean

ii) To get the correct result the data values must be all positive

iii) H.M can be computed from a series with any number of negative q

iv) H.M cannot computed from a series if any of its values are zero

Quartiles: Quartiles are the values which divide the given distribution into 4 equal parts.

Deciles: Deciles are the values which divide the given distribution into 10 equal parts.

Percentiles: Percentiles are the values which divide the given distribution into 100 equal parts.

LECTURE SCHEDULE-3
MEASURES OF DISPERSION

The measures of central tendency serve to locate the centre of the distribution, but they do not reveal how the items are spread out on the either side of the centre. This characteristic of a frequency distribution is commonly referred to as **dispersion**. Dispersion is the measure of the variation of the item.

Small dispersion indicates high uniformity of the items, while large dispersion indicates less uniformity.

The following **measures of variability** (or) **dispersion** are commonly used.

1. Range
2. Standard Deviation
3. Variance
4. Coefficient of Variation

Absolute and Relative Measures of Dispersion:

The dispersion or deviation which gives an idea of spreading or scattering is ascertained as an indication of lack of uniformity. The dispersion may be measured either **absolutely** (or) **relatively**.

When we measure the dispersion and express it in terms of the original data, it is called an **absolute dispersion**. This dispersion is expressed in the unit in which the data are collected.

Absolute measure of dispersion cannot be used for comparison purposes if expressed in different units. For instance, when we measure the height and weight of the students they may be in meters and kilograms respectively – two different units. These different units cannot be measured through absolute method, to know variability.

Types of absolute measure of dispersion.

1. Range
2. Quartile deviation
3. Mean deviation
4. Standard deviation

For comparison of two series having different units of measurement, one requires an unit less measure of dispersion, Such measures are termed as relative measures or coefficient of dispersion.

Type of relative measure of dispersion

1. Co – efficient of variation
2. Co – efficient of Quartile deviation
3. Co – efficient of Mean deviation
4. Co – efficient of Range

Properties (or) Characteristics of a good measure of variation:

Characteristics of an ideal measure of dispersion are the same as that of average, viz., in brief:

1. It should be simple to understand and easy to compute.
2. It should be rigidly defined.
3. It should be based on all observations.
4. It should be amenable to further algebraic treatment.
5. It must have sampling stability.
6. It should not affected by extreme observations

Range

The simplest measure of dispersion is **range**. It is a rough measure of dispersion. The range is the difference between the minimum and maximum values in a group of observations. Its measure depends upon the extreme items and not on all the items. It does not tell us anything about the distribution of values in the series.

Range = Largest value – Smallest value

= **L – S**

Example : Find the 'Range' for the following weights of 15 goats.

30, 25, 14, 42, 18, 26, 21, 11, 35 32, 29, 23, 20, 19, 13.

Range: 49-11=31.

Merits:

i) It is the easiest measure of dispersion,

ii) It can always be found out visually ie., it involves no calculation

Demerits:

i) It is affected by extreme values of the given data

ii) It is not based on all observations

Uses of Range

1. Range is used in industries for the statistical quality control (SQC) of the manufactured product by the construction of Control Chart.
2. Range is useful in studying the variations in the prices of stock, shares and other commodities that are sensitive to price changes from one period to one period.
3. The meteorological department uses the range for weather forecasts since public is interested known the limits within which the temperature is likely to vary on a particular day.
4. In medical, doctors are generally interested in the range of fluctuating temperatures of patients.

Properties

i) It is easy to understand and calculate
ii) It is highly affected by extreme values
iii) It does not depend on all observations
iv) It does not take into account in the form of the distribution

Quartile deviation:

It is the half of the difference between upper and lower quartile values. It is also called semi-inter quartile range. It is not based on all observations. It is not a suitable measure of dispersion to know the variation of the given data.

$$Q.D = \frac{Q_3 - Q_1}{2}$$

Mean deviation:

It is mean of the absolute deviations of observation from its mean. It is based on all observations, but neglecting signs is unjustified and not suitable for algebraic treatment

$$M.D = \frac{\sum |x - \bar{x}|}{n}$$

STANDARD DEVIATION:

It is defined as the positive square root of the mean of the squared deviations from mean. The reason is that it is the square-root of the means of the squared deviation from the arithmetic mean. It provides accurate result. In this method the drawback of ignoring the algebraic sign (in

mean deviation) is overcome by taking the square of deviations, thereby making all the deviations as positive.

Direct Method

Standard Deviation $\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$, $\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$ (**For sample case**)

Short cut method $\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$

Merits, Demerits and uses of Standard Deviation

Merits

1. It is rigidly defined and its value is always definite and based on all the observations and the actual signs of deviations are used.
2. As it is based on arithmetic mean, it has all the merits of arithmetic mean.
3. It is the most important and widely used measures of dispersion.
4. Squaring the deviations makes all of them positive; as such there is no need to ignore the signs (as in mean deviation).
5. It is the basis for measuring the coefficient of correlation, sampling and statistical inferences.
6. The standard deviation provides the unit of measurement for the normal distribution.
7. It can be used to calculate the combined standard deviation of two or more groups.
8. The coefficient of variation is considered to be the most appropriate method for comparing the variability of two or more distributions, and this is based on mean and standard deviation.

Demerits

1. It gives more weight to extreme values, because the values are squared up.
2. It is affected by the value of every item in the series.
3. As it is an absolute measure of variability, it cannot be used for the purpose of comparison.

Uses of Standard Deviation

Standard deviation is the best measures of dispersion. It is widely used in statistics because it possesses most of the characteristics of an **ideal measure of dispersion**. It is

widely used in sampling theory and by biologists. It is used in coefficient of correlation and in the study of symmetrical frequency distribution

Properties of standard deviation

- The S.D of first n natural numbers 1,2 ,…,n is $\sqrt{(n^2-1)/12}$
- The variance or standard deviation is not independent of change of scale
- If all the values of the variable are same, then the standard deviation is zero
- The standard deviation does not alter when a constant quantity k is added to or subtracted from each value of the variable of the series

Variance:

The square of the standard deviation is called the variance.

i.e.) Variance = $(s\tan dard\ deviation)^2 = \sigma^2$

Coefficient of variation:

It is the most important relative measure of dispersion. It is the ratio of S.D to mean expressed as a percentage. It is used to compare the variability of different sets of data having different units of measurement. It is used to know the reliability of data. For example, in an experiment the C.V. % of yield data should be within 5% and 15% .If it is not in this range the data will not be reliable to make conclusion of the experiment.

If we want to compare the variability of two or more series, we can use C.V. The series or groups of data for which the C.V. is greater indicate that the group is more variable, less stable, less uniform, less consistent or less homogeneous. If the C.V. is less, it indicates that the group is less variable or more stable or more uniform or more consistent or more homogeneous.

Formula:

Coefficient of variation (C.V) $= \frac{\sigma}{\bar{x}} \times 100$

Example:

Consider the measurements on yield and plant height of a paddy variety. The mean and SD for yield are 50Kg and 10 Kg respectively. The mean and SD for plant height are 55cm and 5 cm respectively.

Here the measurements for yield and plant height are in different units. Hence the variability can be compared only by using coefficient of variation.

For yield, CV = $\frac{10}{50} \times 100 = 20\%$

For plant height, CV = $\frac{5}{55} \times 100 = 9.1\%$

→ The yield is subject to more variation than the plant height

Properties

i) When the mean value is near to zero the C.V is sensitive to small changes in the mean, limiting its usefulness
ii) The C.V is independent of change of scale but not of origin

Skewness and Kurtosis

Skewness means "lack of symmetry (asymmetrical distribution)". If the frequency distribution uniformly distributed around the mean value then it is said to be symmetrical distribution. If the frequency distribution not uniformly distributed around the mean value is said to be "asymmetrical or skewed one".

There are three types of skewness i.e. positively skewed, negatively skewed and No skewness (or) symmetrical one. It can be determined by using coefficient of skewness. If was given by Karl Pearson.

$$coefficien \quad t\ of \quad skewness \quad = \frac{Mean \quad - Mode}{Stadard \quad Deviation}$$

Drawback of this measure is no limits and also determination of mode position is not possible.

$$coefficien \quad t\ of \quad skewness \quad = \frac{3(Mean \quad - Median \quad)}{S\ \tan \ dard \quad Deviation}$$

Limit of this measure lies between -3 and +3.

Other measure uses the β (read 'beta') coefficient which is given by $\beta_1 = \frac{\mu_3^{\ 2}}{\mu_2^{\ 3}}$ where μ_2 and μ_3 are the second and third central moments. The second central moment u_2 is nothing but variance. The sample estimate of this coefficient is

$$b_1 = \frac{m_3^{\ 2}}{m_2^{\ 3}}$$

Where m_2 and m_3 are the sample central moments given by

$$m_2 = \text{var} \ iance \ = \frac{\sum (x - \overline{x})^2}{n-1} (or) \frac{\sum f(x - \overline{x})^2}{n-1} and \ \ m_3 = \frac{\sum (x - \overline{x})^3}{n-1} (or) \frac{\sum f(x - \overline{x})^3}{n-1}$$

For a symmetrical distribution $b_1=0$. Skewness is positive or regulative depending upon whether m_3 is positive (or) negative.

Kurtosis

Kurtosis is the measure of peaked ness (or) convexity of the curve. Based on peaked ness, kurtosis divided into three types (i) lepto kurtic (ii) Meso kurti and (iii) platy kurtic.

It can be determine based on β_2 values β_2 is nothing but coefficient of kurtosis.

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

Sample estimate of this coefficient is $b_2 = \frac{m^4}{m_2^2}$

Where m_4 is the fourth central moment given by $m_4 = \frac{\sum (x - \overline{x})^4}{n-1} (or) \frac{\sum f(x - \overline{x})^4}{n-1}$

If $b_2 = 3$ then it is said to be meso kurtic, if $b_2$more than 3 then the distribution is called as lepto kurtic. It is less than 3 the distribution is said to be platy kurtic.

It also determine based on $r_2$value $r_2 = \beta_2 - 3$.

If

$r_2 = 0$; meso kurtic

$r_2 > 0$; lepto kurtic

$r_2 < 0$; platy kurtic

LECTURE SCHEDULE-4

PROBABILITY DISTRIBUTIONS

Definition

A probability distribution is a function or rule that assigns probabilities to each value of a random variable

Random Variable

A random variable is a variable that assumes numerical values associated with events of an experiment.

Example

1. Number of patients in a clinic daily is a random variable.
2. Observe 100 babies to be born in a clinic. The number of boys, born is a random variable.
3. Select one student from an university and measure the height and record the height as 'x'. The 'x' is random variable. The value may be from 100 cm to 250 cm

Classification of random variable

A random variable can be classified in to two categories

1. Discrete random variable.
2. Continuous random variable

A discrete random variable is one that can assume only a countable number of values.

A continuous random variable can assume any value in one or more intervals.

In the example above 1st and 2nd example are discrete random variables. The height of the students is a continuous random variable.

Probability Distribution

Suppose ' x ' be a random variable taking countable infinite number of values $x_1 + x_2 +, ...$ with each possible outcomes ' x_i '. We associate a number $p_i = p(x_i)$, called the probability of x_i. the set $[x_i, p(x_i)]$ is called the probability distribution of the random variable ' x '.

The probability distribution can be classified in to two categories

1. Discrete Probability Distribution (or) Probability Mass Function (or) (pmf)
2. Continuous Probability Distribution (or) Probability Density Function (or) (pdf)

Discrete Probability Distribution

Suppose ' x ' is a discrete random variable taking countable infinite number of values $x_1 + x_2 +, ...$ with each possible outcomes ' x_i '. We associate a number $p_i = p(x_i)$, called the probability of x_i.

x	$x_1, x_2, ..., x_n$
$p_i = p(x_i)$	$p_1, p_2, ..., p_n$

Properties

1. $p(x_i) \geq 0$ ie) $0 \leq p(x_i) \leq 1$
2. $\sum_{i=1}^{\infty} p(x_i) = 1$

   This arrangement of probability distribution is called the probability mass function of the random variable ' x '.

Continuous Probability Distribution

The function $f_X(x)$ is known as probability density function (or) simply density function of a random variable and is usually abbreviated as pdf. The expression, $f(x)dx$ can be written as $dF(x)$, is known as the probability differential and the curve $y = f(x)$ is known as the probability density curve (or) simply Probability curve. Hence the probability for a variate value to fall in the finite interval $[a,b]$ is

$P(a \leq x \leq b) = \int_a^b f(x)dx$ which represent the area between the curve $y = f(x)$, the total probability is unity.

ie, $\int_a^b f(x)dx = 1$

where $[a,b]$ is the range of the random variable x. The range may be finite or infinite.

Properties

1. $f(x) \geq 0, -\infty < x < \infty$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. The probability P(E) is given by

$P(E) = \int_E f(x)dx$, is well defined for any event ,E

Mathematical Expectation (or) Expectation

Mean of the probability distribution is called expectation.

Definition

Let x be a random variable taking the values $x_1, x_2, ...x_n$ with the probabilities $p_1, p_2, ..., p_n$ respectively. Then the expectation of x (or) the mathematical expectation is defined by

$$E(x) = x_1 p_1 + x_2 p_2 + ... + x_n p_n$$

$$= \sum_{i=1}^{n} x_i p_i$$

In other words

E(x) = ∑xp(x), if x is discrete random variable.

$E(x) = \int_{-\infty}^{\infty} x.f(x)dx$, if x is continuous random variable

E(x) is also called arithmetic mean

Expectation of a function of a random variable

Consider a random variable x with pdf (or) pmf. If g(x) is a function such that it is a random variable and E[g(x)} is defined as

E[g(x)] = ∑g(x)p(x), if x is discrete random variable.

$E[g(x)] = \int_{-\infty}^{\infty} g(x).f(x)dx$, if x is continuous random variable

Properties

1. E(c) is constant ie) E(c) = c, where c is constant.
2. E(ax ± b) = aE(x) ± b, a and b are constants
3. E(x ±y) =E(x) ± E(y)
4. E(x.y) = E(x).E(y), if x and y are independent.
5. $E(\prod_{i=1}^{n} x_i) = \prod_{i=1}^{n} E(x_i)$
6. E[a .g(x)] = a .E[g(x)]
7. E[g(x)+a] = E[g(x)] + a
8. $V(x) = E(x^2) - [E(x)]^2$

9. $V(ax \pm b) = a^2 v(x)$
10. $V(c) = 0$, c is constant.

Note

If we take $g(x) = x^r$, r being a +ve integer, then,

$$E(x^r) = \int_{-\infty}^{\infty} x^r . f(x) dx$$

which is defined as μ'_r , the rth raw moment of the probability distribution. Thus

$\mu'_r = E(x^r)$

$\mu'_1 = E(x)$

$\mu'_2 = E(x^2)$

Frequency Distribution

Frequency distribution is the distribution of frequencies over different measurements . The frequency distributions are of two types

1. Observed frequency distribution
2. Theoretical frequency distribution

The distributions which are based on actual data or experimentation are called observed frequency distribution. On the other hand, the distribution based on expectation on the basis of past experience is known as theoretical frequency distribution or expected frequency distribution.

For example we toss a coin 200 times, we may get 80 heads and 120 tails, but our expectation is 100 heads and 100 tails. On the basis of the expectation we may test whether the coin is biased or not

Theoretical distributions are

1. Binomial distribution
2. Poisson distribution

} Discrete distribution

3. Normal distribution ____ Continuous distribution

Bernoulli distribution

A random variable x takes two values 0 and 1, with probabilities q and p ie., $p(x=1) = p$ and $p(x=0) = q$, $q = 1 - p$ is called a Bernoulli variate and is said to be Bernoulli distribution where p and q are probability of success and failure. It was given by Swiss mathematician James Bernoulli (1654-1705)

Example

- Tossing a coin(head or tail)
- Germination of seed(germinate or not)

Binomial distribution

The simplest probability distribution is binomial distribution. It is used to represent the probability distribution of discrete random variables.

An experiment which has two possible outcomes is called a Bernoulli trail. The two outcomes are usually called success or failure. An experiment consisting of a repeated number of Bernoulli trails is called a binomial experiment.

Let X is the number of success in 'n' independent Bernoulli trails, with probability 'p' for success in each trail. Then X is called the Binomial random variable with parameters 'n' and 'p'. The binomial distribution is given by the probability mass function, Hence the probability of x success in n trials is given by

$$nC_x p^x q^{n-x}$$

In the formula,

n = number of trials

x = number of successes in a trial

n – x = number of failures in a trial

p = probability of success = x/n

q = 1 – p = probability of failure

Definition

A random variable x is said to follow binomial distribution if it assumes non-negative values and its probability mass function is given by

$$P(X = x) = p(x) = \begin{cases} nC_x p^x q^{n-x} & x = 0,1, ..., n \\ 0 & otherwise \end{cases}$$

The two independent constants n and p in the distribution are known as the parameters of the distribution.

Condition for Binomial distribution

We get the binomial distribution under the following experimentation conditions

1. The number of trial n is finite
2. The trials are independent of each other.
3. The probability of success p is constant for each trial.

4. Each trial must result in a success or failure.
5. The events are discrete events.

Properties

1. If p and q are equal, the given binomial distribution will be symmetrical. If p and q are equal, the distribution will be skewed distribution.
2. Mean = E(x) = np
3. Variance =V(x) = npq (mean>variance)

Application

1. Quality control measures and sampling process in industries to classify items as defectives or non-defective.
2. Medical applications such as success or failure, cure or no-cure.

Poisson distribution

The Poisson distribution, named after Simeon Denis Poisson (1781-1840). Poisson distribution is a discrete distribution. It describes random events that occur rarely over a unit of time or space.

It differs from the binomial distribution in the sense that we count the number of success and number of failures, while in Poisson distribution, the average number of success in given unit of time or space.

Definition

The probability that exactly x events will occur in a given time is as follows

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \quad x=0,1,2\ldots$$ called as probability mass function of Poisson distribution. where λ is the average number of occurrences per unit of time $\lambda = np$

Condition for Poisson distribution

Poisson distribution is the limiting case of binomial distribution under the following assumptions.

1. The number of trials n should be indefinitely large ie., $n \to \infty$
2. The probability of success p for each trial is indefinitely small.
3. $np = \lambda$, should be finite where λ is constant.

Properties

1. Poisson distribution is defined by single parameter λ.

2. Mean = λ
3. Variance = λ. Mean and Variance are equal.

Application

1. It is used in quality control statistics to count the number of defects of an item.
2. In biology, to count the number of bacteria.
3. In determining the number of deaths in a district in a given period, by rare disease.
4. The number of error per page in typed material.
5. The number of plants infected with a particular disease in a plot of field.
6. Number of weeds in particular species in different plots of a field.
7. The poisson distribution applies when (1) the event is something that can be counted in whole numbers; (2) occurrences are independent, so that one occurrence neither diminishes nor increases the chance of another; (3) the average frequency of occurrence for the time period in question is known; and (4) it is possible to count how many events have occurred, such as the number of times a firefly lights up in my garden in a given 5 seconds, some evening, but meaningless to ask how many such events have not occurred. This last point sums up the contrast with the Binomial situation, where the probability of each of two mutually exclusive events (p and q) is known. The Poisson Distribution, so to speak, is the Binomial Distribution Without Q. In those circumstances, and they are surprisingly common, the Poisson Distribution gives the expected frequency profile for events. It may be used in reverse, to test whether a given data set was generated by a random process. If the data fit the Poisson Expectation closely, then there is no strong reason to believe that something other than random occurrence is at work. On the other hand, if the data are lumpy, we look for what might be causing the lump.

LECTURE SCHEDULE-V

Normal distribution

Continuous Probability distribution is normal distribution. It is also known as error law or Normal law or Laplacian law or Gaussian distribution. Many of the sampling distribution like student-t, f distribution and $\chi 2$ distribution.

Definition

A continuous random variable x is said to be a normal distribution with parameters μ and σ^2, if the density function is given by the probability law

$$f\left(x:\mu,\sigma^2\right)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2},\infty<x,\mu<\infty,\ \sigma>0$$

Conditions for Normal Distribution

1. Normal distribution is a limiting forms of the binomial distribution under the following conditions
    - 'n the no. of trials is indefinitely large ie., $n\to\infty$
    - Neither ‘p’ nor ‘q’ is very small.
2. Normal distribution can also be obtained as a limiting form of Poisson distribution with parameter $\lambda\to\infty$
3. Constants of normal distribution are μ and σ^2

Properties of Normal distribution

1. The normal curve is bell shaped and is symmetric at $x=\mu$
2. Mean, median, and mode of the distribution are coincide
   i.e., Mean = Median = Mode = μ
3. It has only one mode at $x=\mu$ i.e., unimodal)
4. The points of inflection are at $x=\mu\pm\sigma$
5. The maximum ordinate occurs at $x=\mu$ and
   its value is $\frac{1}{\sigma\sqrt{2\pi}}$

Standard Normal distribution

A random variable z is said to have a standard normal distribution if its probability function is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-1}{2}z^2}, \quad -\infty < z < \infty$$

A standard normal random variate is denoted by N(0, 1)

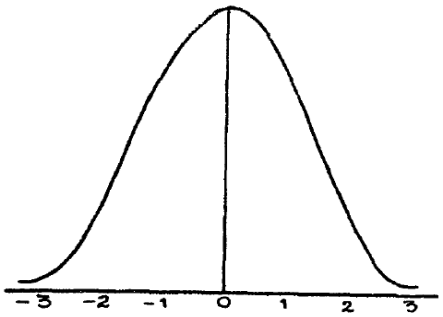


Fig. 9.4. Standard normal curve.

Properties

1. The total area under the curve is 1 sq. unit
2. The maximum ordinate is at mean which is equal to $\frac{1}{\sqrt{2\pi}}$
3. Mean, Median and mode are coincides
4. The ordinate divides the curve into two equal halves each has area 1/2 sq. units

LECTURE SCHEDULE VI

SAMPLING THEORY

Introduction

The basic input in statistical analysis is the data. The data can be collected either by experimentation or by sampling methods.

The sampling methods are used for a variety of purposes.

For example,

i) A consumer wants to purchase a commodity, inspects a portion of it

ii) In industry the quality of a product is assessed through sampling.

iii) In agriculture,

a) to determine the germination rate of a seed material

b) to estimate the area covered under a particular crop

c) to estimate incidence rates of pests and diseases etc.,

Population (Universe)

There is a large collection of objects under investigation we shall call it a population

It can be finite or infinite.

Examples. i) All the students of TNAU

ii) All paddy plants in a field

iii) All the fields under a specified crop

iv) All farms in a village

Finite population

When the number of observation can be counted and is definite, it is known as finite population

- No. of farmers in a village.
- All the fields under a specified crop.

Infinite population

When the number of observation cannot be measured and is infinite, it is known as infinite population.

- Eg – No. of stars in the sky- infinity
- The population of plants on insect in a region.

Parameter

A summary measure that describes any given characteristic of the population is known as parameter. Populations are described in terms of certain measures like mean, standard deviation etc. These measures of the population are called parameter and are usually denoted by Greek letters. For example population is denoted by mean (μ) and variance σ^2.

Sample

A portion or small unit of the total population is considered to study and analysis. Sample consists of a few items of a population.

Example – All the farmers in a village (population) and a few farmers (sample)

Statistic

A summary measure that describes the characteristic of the sample is known as statistic. Thus sample mean, sample standard deviation etc are statistic. The statistic are usually denoted by roman letters.

$\bar{x}$ - sample mean

S – standard deviation

The statistic is a random variable because it varies from sample to sample.

Sampling:

The method of selecting samples from a population is known as sampling.

Sampling technique

There are two ways in which the information is collected during statistical survey. They are

1. Census survey
2. Sampling survey

Sample survey

If the required information is collected through the sample then it is called sample survey.

Census

It is defined as the study of all objects in a population. It is a also known as complete enumeration.

Sample survey Vs Census

Sample survey	Census
The information obtained about the object is less reliable	Reliable

Accuracy of information is not possible	Possible
Is economical in all aspects	Consumes much time energy and money
Is the study of smaller part of population	Study of all objects of population.

Sampling error

The error arising in drawing inference about the population on the basis of few observations (sample) is termed as sampling error.

Need for sampling

The sampling methods have been extensively used for a variety of purposes and in great diversity of situations. Eg. Handful of rice is examined before buying a sack.

In practice it may not be possible to collected information on all units of a population due to various reasons such as

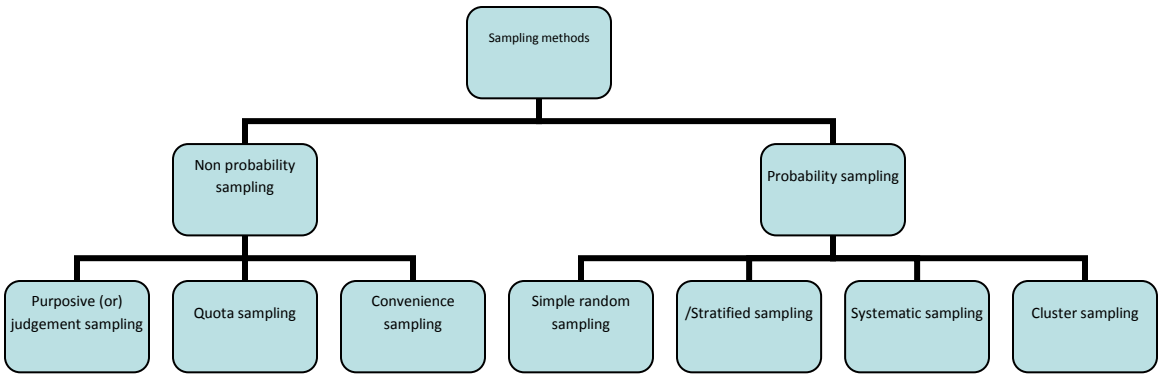
1. Lack of resources in terms of money, personnel and equipment.
2. The experimentation may be destructive in nature. Eg- finding out the germination percentage of seed material or in evaluating the efficiency of an insecticide the experimentation is destructive.
3. The data may be wasteful if they are not collected within a time limit. The census survey will take longer time as compared to the sample survey. Hence for getting quick results sampling is preferred moreover a sample survey will be less costly than complete enumeration.
4. Sampling remains the only way when population contains infinite many numbers.
5. Greater accuracy

Sampling methods

The various methods of sampling can be grouped under

1) Probability sampling or random sampling

2) Non-probability sampling or non-random sampling



Random sampling

Under this method, every unit of the population at any stage has equal chance (or) each unit is drawn with known probability. It helps to estimate the variable of the sample results. It is not possible in non-probability sampling which is used advantageously when there is no frame or when the respondance is expected to be non-co-operative.

Under probability sampling there are two procedures

1. Sampling with replacement(SWR)
2. Sampling without replacement(SWOR)

When the successive draws are made with placing back the units selected in the preceding draws, it is known as sampling with replacement. When such replacement is not made it is known as sampling without replacement.

When the population is finite sampling with replacement is adopted otherwise SWOR is adopted.

Mainly there are many kinds of random sampling. Some of them are.

1. Simple Random Sampling
2. Systematic Random Sampling
3. Stratified Random Sampling
4. Cluster Sampling

1. Unrestricted Sampling

Simple Random sampling (SRS)

It is defined as every item of population is given an equal chance of being included in the sample.. It is used when the population is homogeneous.

A theoretical definition of SRS is as follows

Suppose drawn sample of size n from a population of size N. There are NC_n possible sample of size n. If all possible samples have an equal probability $1/NC_n$ of being drawn, the sampling is said be simple random sampling.

There are two methods in SRS

1. Lottery method
2. Random no. table method

Lottery method

This is most popular method and simplest method. In this method all the items of the universe are numbered on separate slips of paper of same size, shape and color. They are folded and mixed up in a drum or a box or a container. A blindfold selection is made. Required numbers of slips are selected for the desired sample size. The selection of items thus depends on chance.

For example if we select 5 students out of 50 students on slips of the same size and mix them, then we make a blindfold selection of 5 students. This method is also called unrestricted random sampling because units are selected from the population without any restriction. This method is mostly used in lottery draws. If the universe is infinite, this method is inapplicable. There is a lot of possibility of personal prejudice if the size and shape of the slips are not identical.

Random number table method

As the lottery method cannot be used when the population is infinite, the alternative method is using of table of random numbers.

There are several standard tables of random numbers. But the credit for this technique goes to Prof. LHC. Tippet (1927). The random number table consists of 10,400 four-figured numbers. There are various other random numbers. They are fishers and Yates (19380 comprising of 15,000 digits arranged in twos. Kendall and B.B Smith (1939) consisting of 1,00,000 digits grouped in 25,000 sets of 4 digit random numbers, Rand corporation(1955) consisting of 2,00,000 random numbers of 5 digits each etc.,

Ex

Suppose it is required to take a random sample of five plants form a row of 50 plants form the 2 digited random number table the following number are selected, 23, 34, 37, 19, 41. The required sample has 19th, 23rd, 34th, 37th, and 41st plants.

Merits

1. There is less chance for personal bias.

2. Sampling error can be measured.
3. This method is economical as it saves time, money and labour.

Demerits

1. This requires a complete list of the population but such up-to-date lists are not available in many enquires.
2. If the size of the sample is small, then it will not be a representative of the population.

2. Restricted sampling

Stratified Sampling

When the population is heterogeneous with respect to the characteristic in which we are interested, we adopt stratified sampling.

When the heterogeneous population is divided into homogenous sub-population, the sub-populations are called strata. From each stratum a separate sample is selected using simple random sampling. This sampling method is known as stratified sampling.

The number of units to be selected may be uniform in all strata (or) may vary from stratum to stratum.

There are four types of allocation of strata

1. Equal allocation
2. Proportional allocation
3. Neyman's allocation
4. Optimum allocation

If the number of units to be selected is uniform in all strata it is known as equal allocation of samples.

If the number of units to be selected from a stratum is proportional to the size of the stratum, it is known as proportional allocation of samples.

When the cost per unit varies from stratum to stratum, it is known as optimum allocation.

When the costs for different strata are equal, it is known as Neyman's allocation.

Merits

1. It is more representative.
2. It ensures greater accuracy.
3. It is easy to administrate as the universe is sub-divided.

Demerits

1. To divide the population into homogeneous strata, it requires more money, time and statistical experience which is a difficult one.
2. If proper stratification is not done, the sample will have an effect of bias.

Systematic sampling

It is a sampling in which the first member is selected at random and remaining members are selected systematically

Example

Our population size ie., N=50

Our sample size be n=10

$$K = \frac{50}{10} = 5 \text{ units}$$

Select 'r' randomly (ie) r<k

Say r =3

Our units be r, r+k, r+2k, ……r+(n-1)k

3, 8, 13, 18, 23, 28, 23, 33, 38, 43, 48

So we have to select 3rd value, 8th value, 13th value …. 48th value. So we will get a sample of size 10.

Merits

1. This is simple and convenient.
2. The time and work is reduced much.

Demerits

1. It may not represent the whole population.

Cluster Sampling

It is a sampling in which clusters are selected at random and individuals in the selected clusters are studied without omitting even a single individual.

Eg. In a study of fruits of tree, the branches may be numbered and the required number of branches will be selected at random. All the fruits in the selected branches will be studied without omitting even a single fruit. In this sampling different branches are clusters.

Merits

1. It introduces flexibility in the sampling method.
2. It is helpful in large-scale survey where the preparation of the list is difficult, time consuming or expensive.

Demerits

1. It is less accurate than other methods.

Non probability (nonrandom) sampling

Under non probability sampling we have

1) Purposive or Judgement or Deliberate or Directed sampling.
2) Quota sampling.
3) Convenience or Chunk sampling

Purposive or Judgment or Deliberate

The investigator has the power to select or reject any item in an investigation. The choice of sample items depends on the judgment of the investigator. Investigator has the vital role to play in collecting the information.

Merits

1. It is a simple method.
2. It is used to obtain a more representative sample.

Demerits

1. Due to individual bias the sample may not be a representative one.
2. The estimates are not accurate.

Quota sampling

Quota sampling is a type of stratified. The number of units to be selected is decided according to given criteria. But the units are not selected by any random sampling technique. The interviewer is free to any appropriate unit of the population as a sample. In a quota sample, quotas are set up according to some specified characteristics.

It is a stratified-cum-purposive sampling and thus has the advantages of both the method.

Merits

1. The cost per unit selected is considerably less.
2. It is popularly used in market survey and public opinion polls.

Demerits

1. The quotas set are arrived on broad consideration. Detailed study of the population is not made as in proportional allocation.

Convenience sampling

A convenience sample is obtained by selecting convenient population units. The method of convenience sampling is also called the chunk. A chunk refers to that fraction of the

population being investigated which is selected neither by probability nor by judgement but by convenience. A sample obtained from readily available lists such as automobile registrations, telephone directories etc., is a convenience sample and not a random sample even if the sample is drawn at random from the lists.

Hence the results obtained by following convenience sampling method can hardly be representative of the population; they are generally biased and unsatisfactory.

LECTURE SCHEDULE-VII

TEST OF HYPOTHESIS

Test of Significance

It was used to test whether the sample mean equal to population mean (one sample case)(or) whether to test the significant difference between the two sample mean equal to population mean (or) not (two sample case).For testing the above cases if the sample size is large(at least 30 ie.,≥30)apply "z"test (or) "t" test when small sample size(less than 30 ie.,<30)as follows.

Test of Significance

Based on the sampling error the sampling distributions are derived. The observed results are then compared with the expected results on the basis of sampling distribution. If the difference between the observed and expected results is more than specified quantity of the standard error of the statistic, it is said to be significant at a specified probability level. The process up to this stage is known as test of significance

Statistical hypothesis

"In any scientific work, a statement about a phenomenon occurring under certain conditions is formulated as proposition called *scientific hypothesis*".

A scientific hypothesis may be rejected or accepted. For this purpose the scientific hypothesis is translated into statistical language.

Eg. The scientific hypothesis may by "the yield of new paddy variety will be 3500 kg/hec".

In statistical language it may be stated as a random variable (Yield of paddy) is distributed normally with mean (3500 kg/hec). Such a statement or proposition about the probability distribution of random variable is called " a statistical hypothesis".

Simple hypothesis

A hypothesis specifies all parameters of a probability distribution is known as simple hypothesis.

Eg. A statistical hypothesis is involved parameters of a normal distribution, mean and standard deviation called simple hypothesis.

Composite hypothesis

If a hypothesis specifies only some of the parameters of the probability distribution known as composite hypothesis.

Eg. A statistical hypothesis contain, the parameter mean or SD alone called composite hypothesis.

Null hypothesis

The original hypothesis always is not clear and accurate.

The null hypothesis is stated as 'there is no difference between the effect of two treatments or there is no association between two attributes (ie) the two attributes are independent. Null hypothesis is denoted by Ho.

Eg:- There is no significant difference between the two sample means (ie)$\mu 1=\mu 2$.

The hypothesis which reject are accept the original hypothesis is known as Null hypothesis. The original hypothesis known as alternative hypothesis.

Note : The process of deciding whether to accept or reject the null hypothesis is known as testing of hypothesis.

Alternative Hypothesis

The original hypothesis is then stated as an alternative to the null hypothesis. Any hypothesis which is complementary to null hypothesis is called alternative hypothesis, usually denoted by H1.

Eg:- There is a significance difference between the two sample means ie.,

$\mu 1 \neq \mu 2$

$\mu 1 < \mu 2$(one sided alternative-left tailed)

$\mu 1 > \mu 2$(one sided alternative-right tailed)

Level of Significance

The probability levels used for deciding whether to accept or to reject a null hypothesis called levels of significance. The levels of significance is the minimum probability of rejecting, a null hypothesis hypothesis which is in fact true. That is probability of committing Type I error is known as level of significance. Conventionally the levels of significance is taken as 5% or 1%

Critical region: The testing of hypothesis involves the choice of region on the sampling distribution of statistic. If the statistic falls within this region, the null hypothesis is rejected otherwise it is accepted. This region is called the critical region.

One tailed and two tailed test

One tailed test – The critical region will be in one end of the sampling distribution, Hence it is called one tailed test.

Two tailed test – The critical region will be on either end of the sampling distribution. Hence it is two tailed test

Degrees of freedom

The number of degree of freedom is the number of observations that are free to vary after certain restrictions have been placed on the data.

Eg. We are asked to write numbers with restriction imposed that their sum is 30. Then we can choose, only 2 numbers and the third number is automatically determined by relations X1+X2+X3=30 ; X3=30 – (X1+X2), here there are 2 degrees of freedom and the number of restriction is one (ie. The total should be 30)

In general if there are N observations and K restrictions then the degrees of freedom will be N-K.

Steps in Testing of hypothesis

i) Formulation of Null and alternative hypothesis

ii) Specification level of significance

iii) Selection of test statistic and its computation

iv) Find out critical value form tables using the level of significance, sampling distribution and its degree of freedom.

v) Determination of the significance of test statistic

vi) Decision about the null hypothesis based on the significance of the test statistic.

vii) Writing the conclusions in such a way that will answer the question on hand.

Test Statistic

The sampling distribution of a statistic like Z, t, & χ^2 are known as test statistic.

$$\text{Test statistic} = \frac{Statistic - Parameter}{S\tan dardError(Statistc)}$$

Sampling Distribution

The probability distribution is obtained from the frequency distribution. The probability distribution of statistic so obtained is known as sampling distribution of statistic

Testing of Hypothesis

Once the hypothesis is formulated we have to make a decision on it. A statistical procedure by which we decided to accept or reject a statistical hypothesis is called testing of hypothesis.

Sampling Error

From sample data, the statistic is computed and the parameter is estimated through the statistic. The difference between the parameters and the statistic is known as the sampling error.

Decision Errors

By performing a test we make a decision on the hypothesis by accepting or rejecting the null hypothesis Ho. In the process we may make a correct decision on Ho or commit one of two kinds of error.

- We may reject when in fact it is true. This error in decisions is known as Type I error.
- We may accept Ho when in fact it is not true. It is known as Type II error.

	Accept Ho	Reject Ho
Ho is true	Correct Decision	Type I error
Ho is false	Type II error	Correct Decision

The relationship between type I & type II errors is that if one increases the other will decrease.

The probability of type I error is denoted by α. The probability of type II error is denoted by β. The correct decision of rejecting the null hypothesis when it is false is known as the power of the test. The probability of the power is given by $1-\beta$.

Type I and Type II Error

In testing of hypothesis, the decision can be made by accepting or rejecting the null hypothesis. In this decision making, it is possible to commit one of two kind of errors.

i) We may reject null hypothesis when it is correct

ii) We may accept null hypothesis when it is not correct

These errors are known as Type I and Type II errors respectively

Standard Error

The standard deviation of sampling distribution of a statistics is called standard error of the statistic. The square of the standard deviation is known as variance of the statistic.

$$\text{Standard error (S.E)} = \frac{\sigma}{\sqrt{n}}$$

where

σ - population standard deviation.

n - sample size.

* Standard error used to determine the 'limits within which the population parameter may be expected to tie'.

* If also forms the basis of the testing of hypothesis.

Importance of standard error

It plays an important role in the inferential statistics. The reciprocal of the standard error is taken as the measure of precision for statistics. .

LECTURE SCHEDULE-8

LARGE SAMPLE TEST

Large sample test are

1. Sampling from attributes
2. Sampling from variables

Sampling from attributes

There are two types of test for attributes

1. Test for single proportion
2. Test for equality of two proportions

Test for single proportion

In a sample of large size n, we may examine whether the sample would have come from a population having a specified proportion P=Po. For testing

We may proceed as follows

1. **Null Hypothesis (Ho):**
   Ho: The given sample would have come from a population with specified proportion P=Po

2. **Alternative Hypothesis(H_1)**
   H_1 : The given sample may not be from a population with specified proportion

   P≠Po (Two Sided)

   P>Po(One sided-right sided)

   P<Po(One sided-left sided)

3. **Test statistic**

$$Z_o = \frac{|p - P|}{\sqrt{\frac{PQ}{n}}} \sim N(0,1)$$

4. **Level of Significance**
   The level may be fixed at either 5% or 1%

5. **Expected vale**
   The expected value is given by

Z_e = 1.96 at 5% level, 2.58 at 1% level → Two sided test

Z_e = 1.65 at 5% level, 2.33 at 1% level → One sided test

6. **Inference**
   If the observed value of the test statistic Z_o exceeds the table value Z_e we may reject the Null Hypothesis Ho or reject it.

Test for equality of two proportions

Given two sets of sample data of large size n_1 and n_2 from attributes. We may examine whether the two samples are come from the populations having the same proportion. We may proceed as follows

1. **Null Hypothesis (Ho):**
   Ho: The given two sample would have come from a population having the same proportion $P_1=P_2$

2. Alternative Hypothesis (H_1)

H_1 : The given two sample may not be from a population with specified proportion

$P_1 \neq P_2$ (Two Sided)

$P_1 > P_2$(One sided-right sided)

$P_1 < P_2$(One sided-left sided)

3. Test statistic

$$Z_o = \frac{\left| (p_1 - p_2) - (P_1 - P_2) \right|}{\sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}} \sim N(0,1)$$

When $P_1=P_2$ which is not known. Then

$$Z_o = \frac{\left| p_1 - p_2 \right|}{\sqrt{\frac{p_1q_1}{n_1} + \frac{p_2q_2}{n_2}}} \sim N(0,1)$$ for heterogeneous population

Where $q_1 = 1-p_1$ and $q_2 = 1-p_2$

$$Z_o = \frac{\left| p_1 - p_2 \right|}{\sqrt{pq\left(\frac{1}{n} + \frac{1}{n2}\right)}} \sim N(0,1)$$ for homogeneous population

p= combined or pooled estimate.

$$p = \frac{n_1p_1 + n_2p_2}{n_1 + n_2}$$

4. Level of Significance

The level may be fixed at either 5% or 1%

5. Expected vale

The expected value is given by

$Z_e =$ { 1.96 at 5% level; 2.58 at 1% level } → Two sided test

$Z_e =$ { 1.65 at 5% level; 2.33 at 1% level } → One sided test

6. Inference

If the observed value of the test statistic Z_o exceeds the table value Z_e we may reject the Null Hypothesis Ho or reject it.

Sampling from variable

In sampling for variables, the test are as follows

1. Test for single Mean
2. Test for single Standard Deviation
3. Test for equality of two Means
4. Test for equality of two Standard Deviation

Test for single Mean

In a sample of large size n, we examine whether the sample would have come from a population having a specified mean

1. Null Hypothesis (Ho):

Ho: There is no significance difference between the sample mean ie., $\mu=\mu_o$

or

The given sample would have come from a population having a specified mean

ie., $\mu=\mu_o$

2. Alternative Hypothesis(H_1)

H_1 : There is significance difference between the sample mean ie., $\mu=\mu_o$

ie., $\mu\neq\mu_o$ or $\mu>\mu_o$ or $\mu<\mu_o$

3. Test statistic

$$Z_o = \frac{\left|\bar{x} - \mu\right|}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

When population variance is not known, it may be replaced by its estimate

$$Z_o = \frac{\left|\bar{x} - \mu\right|}{\frac{s}{\sqrt{n}}} \sim N(0,1)$$

Where $s = \sqrt{\frac{\sum x^2 - \frac{\left(\sum x\right)^2}{n}}{n-1}}$

4. Level of Significance

The level may be fixed at either 5% or 1%

5. Expected vale

The expected value is given by

Z_e = 1.96 at 5% level, 2.58 at 1% level → Two sided test

Z_e = 1.65 at 5% level, 2.33 at 1% level → One sided test

6. Inference

If the observed value of the test statistic Z_o exceeds the table value Z_e we may reject the Null Hypothesis Ho or reject it.

Test for equality of two Mean

Given two sets of sample data of large size n_1 and n_2 from variables. We may examine whether the two samples are come from the populations having the same mean. We may proceed as follows

1. Null Hypothesis (Ho):

Ho: There is no significance difference between the sample mean ie., $\mu=\mu_o$

or

The given sample would have come from a population having a specified mean

ie., $\mu_1=\mu_2$

2. Alternative Hypothesis(H_1)

H_1 : There is significance difference between the sample mean ie., $\mu=\mu_o$

ie., $\mu_1\neq\mu_2$ or $\mu_1<\mu_2$ or $\mu_1>\mu_2$

3.Test statistic

$$Z_o = \frac{\left|(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)\right|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma^2{}_2}{n_2}}} \sim N(0,1)$$

When population variance known, $\sigma_1= \sigma_2 = \sigma$

$$Z_o = \frac{\left|(\bar{x}_1 - \bar{x}_2)\right|}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0,1)$$

When population variance is unknown. They may be replaced by their estimates s_1^2 and s_2^2

$$Z_o = \frac{\left|(\bar{x}_1 - \bar{x}_2)\right|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0,1)$$ Two samples are come from different population.

When σ_1^2 and σ_2^2 are unknown. They may be replaced by combined estimate or pooled estimate

$$Z_o = \frac{\left|(\bar{x}_1 - \bar{x}_2)\right|}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0,1)$$ Two samples are come from the sample population

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$$

4. Level of Significance

The level may be fixed at either 5% or 1%

6. Expected vale

The expected value is given by

$Z_e =$ { 1.96 at 5% level, 2.58 at 1% level } → Two sided test

$Z_e =$ { 1.65 at 5% level, 2.33 at 1% level } → One sided test

6. Inference

If the observed value of the test statistic Z_o exceeds the table value Z_e we may reject the Null Hypothesis Ho or reject it.

LECTURE SCHEDULE -XIV

SMALL SAMPLE TEST

Sampling problems discussed so far dealt with means and proportions. Evaluation of their sampling errors was based on the normal distribution. In the case of the mean, the sampling distribution was normal because the variable was distributed normally in the population or because the Central Limit Theorem ensured normality for large samples. In the case of proportions, the normal distribution was used as an approximation for the underlying binomial distribution. When samples are small, $n < 30$ when the population is normally distributed, and when the population variance has to be estimated from sample data, the distribution of the sample mean is no longer normal. A small sample distribution, known as the t-distribution, has to be used in this case. When samples are small and the distribution of the variable in the population is not normal, there is no readily available sampling distribution.

When n less than 30 we apply small sample test. The tests are

1. Test based on student t distribution $-\infty \leq t \leq \infty$
2. Test based on F distribution $0 \leq F \leq \infty$
3. Test based on χ^2 distribution $0 \leq \chi^2 \leq \infty$

Test based on student t - Distribution

t – Distribution is used to test.

1. Test for single mean for single sample case.
2. Test for equality of two means for double sample case (independent samples and dependent samples (paired – t- test)
3. Test for significance of observed correlation coefficient.
4. Test for significance of observed partial and multiple correlation coefficients.
   Test for significance of observed regression coefficient

Properties of t – test

1. The variable t-distribution ranges from $-\infty$ to $+\infty$.
2. Like standard normal distribution, the t – distribution is symmetrical and has a mean zero.
3. Variance of the t-distribution is greater than one.

Assumption about t-test

1. The random variable X ~ Normal distributions or the sample is drawn from a normal population.
2. All observation in the sample are independent
3. The sample size is small ie less than 30. Also the sample should not contain less than 5 observations.

Definition of student t distribution

The ratio, Z= $\frac{|\bar{x} - \mu|}{\sigma / \sqrt{n}}$ is a standard normal variable when ' σ' is unknown , it can be substituted by the sample SD 's' , with small samples, the ratio.

$$t = \frac{|\bar{x} - \mu|}{s / \sqrt{n}}$$

This fact was brought out by Sir William Gosset under the pen name 'student'. Hence this ratio is called student t.

Tests of significance for small samples ('t' test)

One sample test

Assumption

- Sample is drawn at random
- Samples are independent in nature

Conditional

- Sample is small
- σ not known

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

where $s = \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$ t-distribution with $_{(n-1)df}$

F distribution

F-test (or) variance Ratio test – The F-test is named by R.A. Fisher

Assumption

1) Normality – values in each group are normally distributed.
2) Homogeneity – Variance within each group should be equal for all groups ($\sigma_1^2=\sigma_2^2=\ldots=\sigma_c^2$)
3) Independence of error – It states that the error should be independent for each other.

Test based on F – distribution

Given two sets of identically independently distributed observations sizes n_1 and n_2 from normal population, we may define the χ^2 variates χ_1^2 and χ_2^2 with $\nu_1=n_1-1$ and $\nu_2 = n_2 -1$ from the two samples respectively. The ratio of two χ^2 variates is given by

F- Statistic

$$F = \frac{\chi_1^2 / v_1}{\chi_2^2 / v_2} = \frac{\chi_1^2 / n_{i-1}}{\chi_2^2 / n_{1-1}}$$

F – Statistic is used to test.

1. Significance of equality of two variances.
2. Significance of observed multiple correlation coefficient.
3. Significance of observed correlation ratio.
4. Equality of several means of normal population in several sample case. This is known as the Analysis of variance Technique (ANOVA) applied in design of experiments.

Test for equality of variances

$$F = \frac{S_1^2}{S_2^2} \sim F(n_1 - 1, n_2 - 1) \quad \text{if } S_1^2 > S_2^2$$

$$F = \frac{S_2^2}{S_1^2} \sim F(n_2 - 1, n_1 - 1) \quad \text{if } S_2^2 > S_1^2$$

Where

$$S_1^2 = \frac{\sum x_1^2 - \frac{\left(\sum x_1\right)^2}{n_1}}{n_1 - 1}$$

$$S_2^2 = \frac{\sum x_2^2 - \frac{\left(\sum x_2\right)^2}{n_2}}{n_2 - 1}$$

Test for equality of two means: - (Independent samples)

i) For testing the equality of two mean we have to test the variance using F-test, variance are equal and proceed t test using the formula given below

Variances are equal

$$t = \frac{\left|x_1 - x_2\right|}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{(n1+n2-2)df}$$

$$S^2 = \frac{SS(1) + SS(2)}{n_1 + n_2 - 2}$$

$$SS(1) = \sum x_1^{\,2} - \frac{\left(\sum x_1\right)^2}{n_1}$$

$$SS(2) = \sum x_2^{\,2} - \frac{\left(\sum x_2\right)^2}{n_2}$$

ii) For testing the equality of two mean we have to test the variance using F-test, variance are unequal(sample size are equal ie., $n_1=n_2$) and proceed t test using the formula given below

Variances are unequal

$$t = \frac{\left|x_1 - x_2\right|}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{\left(\frac{n1+n2}{2}-1\right)} df$$

$$S^2 = \frac{SS(1) + SS(2)}{n_1 + n_2 - 2}$$

iii) For testing the equality of two mean we have to test the variance using F-test, variance are unequal(sample size are equal ie., $n_1 \neq n_2$) and proceed t test using the formula given below

Variances are unequal

$$t = \frac{\left|x_1 - x_2\right|}{\sqrt{\frac{S_1^{\,2}}{n_1} + \frac{S_2^{\,2}}{n_2}}} \sim t_w df \text{ (Bahrain's Fisher d- Distribution)}$$

Where $t_w = \dfrac{t_1\left(\frac{S_1^{\,2}}{n_1}\right) + t_2\left(\frac{S_2^{\,2}}{n_2}\right)}{\frac{S_1^{\,2}}{n_1} + \frac{S_2^{\,2}}{n_2}}$

Where t_1 = t with (n_1-1) df

t_2 = t with (n_2-1) df

Test for equality of two means: (Dependent samples or Paired t – test)

Two samples are said to be dependent when the elements in one sample are related to those in the other in any significant or meaningful manner. The two samples may consist of pairs

of observations made on same object. In double sample case, for testing the significance difference of the means, we may have the conditions that,

1. The sample sizes are equal. ie) $n_1 = n_2 = n$.
2. The sample observations refer to the characteristics of the same set of objects.

In such a case, the two samples can be combined together pairwise as (x_i, y_i), i= 1.2……..n of size 'n' and the difference of each pair $d_i = x_i - y_i$, i=1,2….n may be obtained. Now we have a single sample of size 'n', for the variables.

For testing the equality of the means we may use the test statistic,

$$t = \frac{\bar{d}}{S/\sqrt{n}} \sim t_{(n-1)}\, df$$

where $d_i = x_i - y_i$

$$\bar{d} = \frac{\sum d_i}{n} \quad \text{here } n_1 = n_2 = n$$

$$S = \sqrt{\frac{1}{n-1}\left[\sum d_i^{\,2} - \frac{\left(\sum d_i\right)^2}{n}\right]}$$

Chi-square test

It is used to test

i) the independence of attributes
ii) the goodness of fit
iii) the homogeneity of variances
iv) the homogeneity of correlation coefficients
v) the linkage in genetic problems

Contingency table: A two-way table with two attributes is known as contingency table.

For example, an entomologist may be interested to know the effectiveness of different concentrations of a chemical in killing the insects. In this case the concentrations of the chemical form one attribute. The state of insects 'killed' and 'not killed' forms another attribute. The results of this experiment can be arranged in the form of a contingency table. The association between these two attributes may be tested using χ^2 statistic.

Goodness of Fit

In testing of hypothesis our objective may be to test whether a sample has come from a population which has a specified theoretical distribution like normal, binomial and Poisson. In order words, it may be necessary to test whether an obtained frequency distribution resembles a theoretical distribution. In plant genetics, our interest may be to test whether the observed segregation ratios deviate significantly from the Mendelian ratios. In such situations we want to test agreement between the observed and theoretical frequencies. Such test is called as test of goodness of fit.

The null hypothesis in goodness of fit tests is that there is no disagreement between the observed and theoretical distributions, or the observed distribution fits to the theoretical distribution. The test statistic to test this null hypothesis is

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where O = observed frequency,
E = expected frequency

$$Expected\ \ frequency = \frac{Corresponding\ row\ total \times Corresponding\ column\ total}{Grand\ Total}$$

In general, the degrees of freedom is associated with this χ^2 is k – 1, where k is the number of classes or groups.

2X2 Contingency table

When the individuals are classified into two categories with respect to each of the two attributes then the table showing frequencies distributed over 2x2 classes is called 2x2 contingency table

Example: Suppose the individuals are classified according to two attributes colour (B) and intelligence (A). The distribution of frequencies over cells is shown in table 1

Table

	A1	**A2**	**Total**
B1	a	b	a+b=R1
B2	c	d	c+d=R2
Total	a+c=C1	c+d=C2	a+b+c+d=N

where a, b, c and d are frequencies of different cells

Formula:

$$\chi^2 = \frac{N(ad-bc)^2}{R_1 R_2 C_1 C_2} \text{ with 1 d.f}$$

rxc Contingency Table

Definition: A contingency table is a tabular arrangement of count data representing how the row factor frequencies relate to the column factor. We call a contingency table with "r" rows and "c" columns, an r x c contingency table. Each category in a contingency table is called a cell.

Purpose: comparing percentages or testing of association.

General Model for an $r \times c$ Contingency Table

		Column variable					**Row sums**
		C_1	C_2	...	C_j	... C_c	
Row variable	R_1	O_{11}	O_{12}	...	O_{1j}	... O_{1c}	$O_{1.}$
	R_2	O_{21}	O_{22}	...	O_{2j}	... O_{2c}	$O_{2.}$
	⋮	⋮	⋮		⋮	⋮	⋮
	R_i	O_{i1}	O_{i2}	...	O_{ij}	... O_{ic}	$O_{i.}$
	⋮	⋮	⋮		⋮	⋮	⋮
	R_r	O_{r1}	O_{r2}	...	O_{rj}	... O_{rc}	$O_{r.}$
Column sums		$O_{.1}$	$O_{.2}$	...	$O_{.j}$	... $O_{.c}$	n

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where O = observed frequency,
E = expected frequency

$$Expected\ \ frequency = \frac{Corresponding\ row\ total \times Corresponding\ column\ total}{Grand\ Total}$$

In general, the degrees of freedom associated with this χ^2 is (r-1)(c-1), where r, c are rows and columns.

Yates correction for continuity

In the 2x2 contingency table, if the expected cell frequencies are large, the discrete distribution of the probabilities of the cell frequencies approximate to normal distribution and hence χ^2-statistics is distributed as χ^2 distribution with 1 d.f. On the other hand, if the expected frequencies are small(less than 5) the distribution of χ^2 cannot be used

The expected value of χ^2 can also be obtained directly using the expression

.

$$\chi^2 = \frac{N\left(|ad-bc| - \frac{N}{2}\right)^2}{R_1 R_2 C_1 C_2}$$

LECTURE SCHEDULE X

CORRELATION

If the change of one variable affects the change of another variable then the variables are said to be correlated. In other words, the systematic interrelationship between the variables is termed as correlation

Eg. i) Price and demand
ii) Population and unemployment
iii) Yield and pest incidence

Types of Correlation

1. Simple correlation
2. Simple linear correlation
3. Simple Non-linear correlation
4. Multiple correlation
5. Multiple linear correlation
6. Multiple Non-linear correlation
7. Positive or Direct Correlation
8. Negative or Indirect Correlation
9. No Correlation or Independent variables
10. Pseudo Correlation

Simple correlation

If the relationship between only two variables x and y then the correlation is said to be Simple correlation.

Simple linear correlation

If the relationships between two variables are straight line then the correlation is said to be Simple linear correlation.

Simple Non-linear correlation

If the relationships between two variables are not straight line then the correlation is said to be Simple Non-linear correlation

Multiple correlation

If the relationship between more than two variables then the correlation is said to be Multiple correlation.

Multiple linear correlation

If the relationships between more than two variables are in straight line then the correlation is said to be multiple linear correlation.

Multiple Non-linear correlation

If the relationships between more than two variables are not in straight line then the correlation is said to be Multiple Non-linear correlation.

Positive correlation

An increase in one variable may cause an increase in other variable, or a decrease in one variable may cause a decrease in the other variable. In other words if the movement of the variables are in the same direction then they are said to be positively correlated. It is also called direct correlation.

Eg. i) Yield and fertilizer application
ii) Changes in the temperature and changes in length of a iron bar
iii) Increase in the radius of the circle and increase in the circumference.
iv) Incidence of pests and weather factors

Negative correlation

It the movements of the variable are in opposite direction then the variables are said to be negatively correlated. It is also called indirect correlation

i) Yield and pest incidence

No Correlation or Independent Variables

If the variables are having no relationship then the variables are said to be independent variables or no correlated one.

SCATTER DIAGRAM

In correlation problems, first we have to investigate whether there is any relation between the variables say, X and Y. For this purpose we use scatter diagram.

Let (X_1, Y_1), (X_2, Y_2), (X_3, Y_3) . . . . . (X_n, Y_n) be n pairs of obsevations. If the values of variables X and Y are plotted along the X axis and Y axis respectively in the XY plane of graph sheet, the resultant diagram of dots is known as scatter diagram. From the scatter diagram we can say whether there is any correlation between X and Y, correlation is positive or negative and the correlation is linear or curve linear.

Advantages:

1. Scatter diagram method is easy to understand and simple to follow.
2. It does not involve too much of mathematics.

1. It helps to get a preliminary idea of correlation in a data.

Disadvantages:

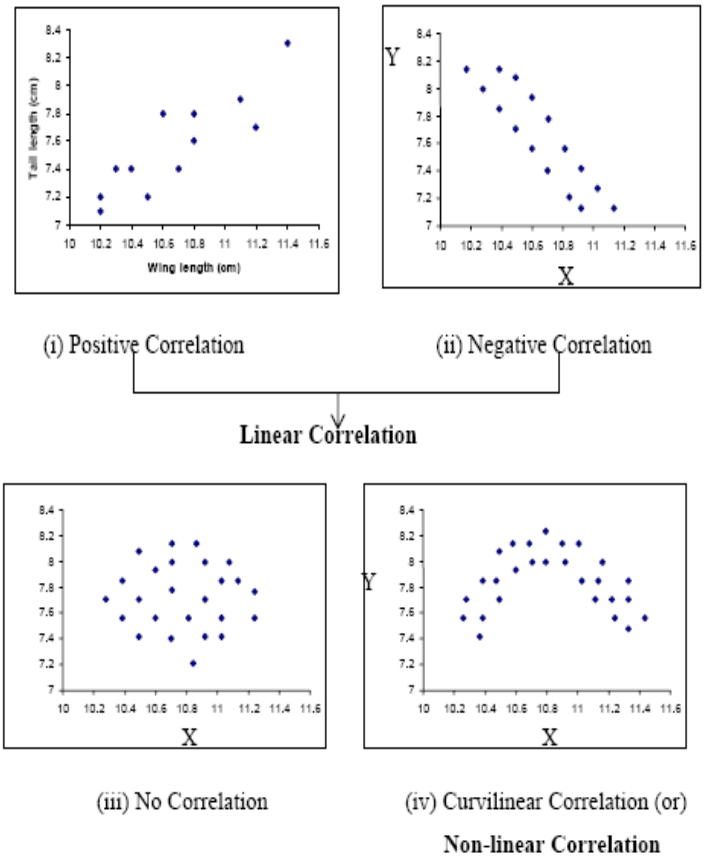
1. It gives only the direction but not magnitude of correlation between the two variables.

Example

The following data stands for wing and tail lengths among birds of a particular species. Draw the scatter diagram.

Birds	1	2	3	4	5	6	7	8	9	10	11	12
Wing length (cm) x:	10.4	10.8	11.1	10.2	10.3	10.2	10.7	10.5	10.8	11.2	10.6	11.4
Tail length (cm) y:	7.4	7.6	7.9	7.2	7.4	7.1	7.4	7.2	7.8	7.7	7.8	8.3

Scatter Diagram



(i) Positive Correlation (ii) Negative Correlation

Linear Correlation

(iii) No Correlation (iv) Curvilinear Correlation (or)

Non-linear Correlation

Correlation coefficient

The scatter diagram will give only a vague idea about the presence or absence of correlation and the nature of the correlation. It will not indicate about the strength or degree or relationship between two variables. The index of the degree of relationship between two variables is known as correlation coefficient. It can be determined by the following formula

$$r = \frac{SPxy}{\sqrt{SSx.SSy}}$$

where, $SP_{xy} = [\Sigma xy - (\Sigma x \Sigma y) / n]$

$SS_x = [\Sigma x^2 - (\Sigma x)^2 / n)]$

$SS_y = [\Sigma y^2 - (\Sigma y)^2 / n)]$

The two variables X & Y are said to be positively, negatively correlated according as r is positive or negative. If r is equal to zero then the two variables are said to be uncorrelated.

Properties of correlation co-efficient

i) 'r' will always lie between **– 1 & +1**

ii) The value of 'r' is unaltered if X & Y are interchanged. This property is called **symmetric property.**

iii) Suppose the values of data on two variables are altered uniformly then the value of 'r' will not be altered. This property is called **invariance property of 'r'**.

Limitation of correlation co-efficient

1. If the variable under study should be continuous and normally distributed one.
2. Variables are linear in nature.
3. Each pair of observation unconnected with other pairs of observations.

Spearman's Rank Correlation

Sometimes we come across statistical series in which the variable under consideration are not capable of quantitative measurements but can be arranged in a serial order. This happens when we are dealing with qualitative characteristics (attributes) such as honesty, beauty, character, morality, etc. This cannot be measured quantitatively but can be arranged serially. In such situations Karl Pearson's coefficient of correlation cannot be used as such. Charles Edward Spearman, a British psychologist, developed a formula in 1904 which consists in obtaining the correlation coefficient between the ranks of n individuals in the two attributes under study.

The Pearson Correlation Coefficient between the ranks X and Y is called the rank correlation coefficient between the characteristics A and B for that group of individuals.

The students are assigned ranks in Statistics according to their marks in Statistics. Also, they are assigned ranks in Mathematics according to their marks in Mathematics. Then, the correlation between these two sets of ranks is called rank correlation. The coefficient of correlation computed for these ranks is called Spearman's coefficient of rank correlation.

In a bivariate data, if the values of the variables are ranked in the decreasing (or increasing) order, the correlation between these ranks is rank correlation. The coefficient of correlation computed for these rank is Spearman's coefficient of rank correlation. It is denoted by ρ (Rho)

If R1 and R2 are the ranks in the two characteristics, and d = R1 – R2 is the difference between the ranks, coefficient of rank correlation is –

$$\rho = 1 - \frac{6\sum d_i^2}{n^3 - n}$$ (When ranks are not repeated)

Where n is the number of observations

Since ρ is the product moment coefficient of correlation between the ranks, it is a value between -1 and +1

Karl Pearson's coefficient of correlation can be calculated only if the characteristics under study are quantitative (they should be numerically measurable) but, Spearman's coefficient of rank correlation can be calculated even if the characteristics under study are qualitative. If it is possible to assign ranks to the units with regard to the two characteristics, co efficient of rank correlation can be calculated.

REPEATED RANKS

In case of attributes if there is a tie i.e. if any two or more individuals are placed together in any classification w.r.t an attribute or if in case of variable data there is more than one item with the same value in either or both the series, then Spearman's formula for calculating the rank correlation coefficient breaks down, since in this case the variable X (the ranks of individuals in characteristic A (1st series) and Y (the ranks of individuals characteristic B (2nd series) do not take the values from 1 to n and consequently $x \neq y$, while in proving we had assumed that $x = y$.

For the computation of coefficient of rank correlation, while ranking the values, two or more values may be equal. And so, a situation of ties may arise. In such a case, all those values which are equal are assigned with the **same average rank**. And then, the coefficient of rank correlation is found. Here, corresponding to every such repeated rank correlation is found. Here corresponding to every such repeated rank (which repeats m times), a factor $(m^3 - m) / 12$ is added to $\sum d^2$

In this case, common ranks are assigned to the repeated items. These common ranks are the arithmetic mean of the ranks which these items would have got if they were different from, each other and the next item will get the rank next to the rank used in computing the common rank. For e.g, suppose an item is repeated at rank 4. Then the common rank to be assigned to each item is (4 + 5) / 2 i.e, 4.5 which is the average of 4 and 5 , the ranks which these observations would have assumed if they were different. The next item will be assigned the rank 6. if an item is repeated thrice at rank 7, then the common rank to be assigned to each value will be (7+8+9)/ 3, i.e 8 which the arithmetic mean of 7,8 and 9 viz, the ranks these observation would have got if they were different from each other. The next rank to be assigned will be 10.

If only a small proportion of the ranks are tied, this technique may be applied together with formula. If a large proportion of ranks are tied, it is advisable to apply an adjustment or a correction factor as explained: "In a formula add the factor $\frac{m^3 - m}{12}$ to $\sum d_i^2$ where m is the number of times an item is repeated. This correction factor is to be added for each repeated value in both the series.

Formula:

When the ranks are repeated: $\rho = 1 - \frac{6\left(\sum d_i^2 + \frac{1}{12}\left(m_1^3 - m_1\right) + \frac{1}{12}\left(m_{21}^3 - m_2\right) + \ldots \right)}{n^3 - n}$

ADVANTAGES OF SPEARMAN'S RANK CORRELATION COEFFICIENT

1. Since Spearman's rank correlation coefficient ρ is nothing but Pearson's correlation coefficient between the ranks, it can be interpreted in the same way as the Karl Pearson's correlation coefficient.
2. Karl Pearson's correlation coefficient assumes that the parent population from which sample observations are drawn is normal. If this assumption is violated than we need a measure which is distribution – free (or non – parametric). A distribution free measure is one which does not make any assumptions about the form of the population. Spearman's ρ is such a measure (i.e. distribution free), since no strict assumptions are made about the form of the population from which sample observations are drawn.
3. Spearman's formula is easy to understand and apply as compared with Karl Pearson's formula. The values obtained by the two formulae, viz Pearsonian r and Spearman's ρ are generally different. The differences arise due to the fact that when ranking is used instead of full set of observations, there is always some loss of information. Unless many ties exist, the coefficient of rank correlation should be slightly lower than the Pearsonian coefficient.
4. Spearman's formula is the only formula to be used for finding correlation coefficient if we are dealing with qualitative characteristics which cannot be measured quantitatively but can be arranged serially. It can also be used where actual data are given. In case of extreme observations, Spearman's formula is preferred to Pearson's formula.
5. Spearman's formula has its limitation also. It is not practicable in the case of bivariate frequency distribution. For n > 30, this formula should not be used unless the ranks are given, since in the contrary case the calculations are quite time consuming.

LECTURE SCHEDULE-XI
REGRESSION

The concept of regression was given by **Francis Galton**. Of two variables under study one may represent the cause and the other may represent the effect. The variable representing the cause is known as independent variable and it is denoted by X. The variable X is sometimes called predictor variable or regressor. The variable representing the effect is known as dependent variable and is denoted by Y. The variable Y is sometimes called predicted variable. The relationship between the independent and dependent variables may be expressed as a function. Such functional relationship between two variables is termed as regression.

When only two variables are involved the functional relationship is known as **simple regression**. If the relationships between the two variables are straight line then it is known as **simple linear regression**. Otherwise it is called as **simple nonlinear regression**. When there are more than two variables and one of them is assumed to be dependent upon the others, the functional relationship between the variables is known as **multiple regressions.**

Curve fitting

In general when two variables are studied simultaneously the interest will be to know the functional relationship between them. The scatter diagram is used to identify the type of relationship between the two variables for example, when n pairs of observations (X_1, Y_1), $(X_{12}$ $Y_{2)} \ldots . (X_n, Y_n)$ are plotted on a graph sheet.

The scatter diagram for a given data may result in non – linear forms. The functional relationships may be given by algebraic expressions like

$$Y = a + bX$$

The above algebraic expression is known as curve. The method of finding such relationship is known as curve fitting. A best fitting line is one for which the sum of the squares of the residuals, (or errors) is minimum. For this purpose the principle, known as method of least squares is used.

Method of least square

A best fitting line is one for which sum of the squares of the residuals (errors) is minimum. The principle in which minimize the error is known as method of least squares.

General form of simple linear equation is

$$y = a + bx$$

then estimated values of y is

$$\hat{y} = a + bx$$

Let

$$y-\hat{y} = e_i$$

Application of methods of least squares principle

$$\sum e_i^2 = \text{minimum}$$
$$\sum (y - \hat{y})^2 = 0$$
$$\sum [y -(a+bx)]^2 = 0$$
$$\sum (y -a-bx)^2 = 0$$
$$s = \sum (y-a-bx)^2 = 0$$

For Minimization

$$\frac{\partial s}{\partial a} = 2 \sum (y-a-bx)(-1)$$
$$\frac{\partial s}{\partial a} = 0$$
$$-2\sum (y-a-bx) = 0$$
$$\sum (y-a-bx) = 0$$
$$\sum y-\sum a-\sum bx=0$$
$$\sum y-na-b\sum x=0 \qquad (1)$$

Note $\sum (c) = nc$, $\sum ax = a\sum x$

Similarly

$$\frac{\partial s}{\partial b} = 2 \sum (y - a - bx)(-x)$$
$$\frac{\partial s}{\partial b} = 0$$
$$-2 \sum (y-a-bx)(x)=0$$

$\sum$ (y –a – bx) (x)=0

$\sum$ (xy-ax-bx^2) =0

$$\sum \text{xy-a}\sum \text{x-b}\sum \text{x}^2 = 0 \qquad (2)$$

Equation (1) and (2) are normal equations

From normal equation (1)

$$\sum \text{y} - \text{na} - \text{b}\sum \text{x} = 0$$

$\div n$

$$\frac{\sum y}{n} - \frac{na}{n} - \frac{b\sum x}{n} = 0$$

$$\frac{\sum y}{n} - a - b\frac{\sum x}{n} = 0$$

$$a = \frac{\sum y}{n} - b\frac{\sum x}{n}$$

[redacted]

From Normal Equation (2)

$$\sum \text{xy} - \text{a}\sum \text{x} - \text{b}\sum \text{x}^2 = 0$$

Substitute a = $\frac{\sum y}{n} - b\frac{\sum x}{n}$ *then*

$$\sum xy - \left[\frac{\sum y}{n} - \frac{b\sum x}{n}\right]\sum x - b\sum x^2 = 0$$

$$\sum xy - \left[\frac{\sum x \sum y}{n} - \frac{b(\sum x)^2}{n}\right] - b\sum x^2 = 0$$

$$\sum xy - \frac{\sum x \sum y}{n} + \frac{b(\sum x)^2}{n} - b\sum x^2 = 0$$

$$\sum xy - \frac{\sum x \sum y}{n} = b\left[\sum x^2 - \frac{(\sum x)^2}{n}\right]$$

[redacted]

Substitute the value of a and b to the general form of simple linear regression equation, then that equation y = a+bx is said to be fitted simple linear regression equation y on x.

Regression coefficient

In the equation Y = a +bX, 'b' is the slope of the line, also called Regression coefficient and 'a' is the intercept of the line with the Y-axis.

b_{YX} = regression coefficient Y on X

b_{XY} = regression coefficient X on Y

Properties of regression coefficients

i) Correlation coefficient is the geometric mean between the regression coefficient
ii) If one of the regression Coefficients is greater than unity, the other must be less than unity.
iii) Arithmetic mean of regression coefficients is greater than the correlation coefficient.
iv) Regression coefficient are independent of the origin but not the scale

Uses of Regression:

The regression analysis is useful in predicting the value of one variable from the given value of another variable. Such predictions are useful when it is very difficult or expensive to measure the dependent variable.

The other use of the regression analysis is to find out the causal relationship between variables. Suppose we manipulate the variable X and obtain a significant regression of variables Y on the variable X. Thus we can say that there is a causal relationship between the variable X and Y.

The causal relationship between nitrogen content of soil and growth rate in a plant, or the dose of an insecticide and mortality of the insect population may be established in this way.

Experimental Designs

Choice of treatments, method of assigning treatments to experimental units and arrangement of experimental units in different patterns are known as designing an experiment. We study the effect of changes in one variable on another variable. For example how the application of various doses of fertilizer affects the grain yield. Variable whose change we wish to study is known as response variable. Variable whose effect on the response variable we wish to study is known as factor.

Any researcher before conducting an experiment has to plan and design the experiment. While designing an experiment, the researcher is allowed to study the factors of his interest and show that these factors actually cause certain effects. An experiment has to be designed by taking into consideration the following aspects, viz., objective of the study, experimental material availability, cost, time etc. Prof. R. A .Fisher has developed modern concepts of experimental design in the planning and conduct of agricultural field experiments. They are now used in all fields of science. In agricultural field experiments, an experiment may be conducted to see the effect of different fertilizers. In this situation, the field has to be divided into a number of plots of equal size and same shape, homogeneous in respect of soil fertility, irrigation, crop, seed, plant protection umbrella, crop management etc. and each plot receives the fertilizer treatment randomly. The variability in crop yield (or) any other criteria will enable the investigator to distinguish between fertilizers. The different terms used are:

Definition

The logical construction of the experiment in which the degree of uncertainty with which the inference is drawn may be well defined or Is the layout of a detailed experimental plan in advance of doing the experiment.

Experiment: An experiment is just a test or series of test

Experimental Unit

A Subject (or) a group of object (or) the total material to which a treatment is applied in a trial in a single replication is known as an Experimental unit.

Blocks

In agricultural experiments we divide the whole experimental unit(field) into homogeneous sub groups or strata. These strata which are uniform among themselves than the field as a whole are known as blocks.

Factor is another name for the independent variables of an experimental design.

Treatment: Various object of comparison in a comparative experiment are termed as treatment

Ex: Application of fertilizers to field plots

Experiment unit: Is the object to which treatment is applied to record the observations

Ex: a group of insects in entomological experiments

Experimental error

The error caused by the extraneous factors that are beyond the control of human approach is known as experimental error.

For instance germination of seed, plant growth numbering of pods (or) ears per plant, response to feed by the animal, taste of a person etc. are the factors beyond human control.

The reduction Experimental error provides smaller standard error for a treatment means or for the difference between two treatments means which results in detecting small difference between treatment means. In this way the efficiency of the experiment increases.

Extraneous factors:

In agricultural field experiments, the variation in responses is caused by number of factors like heterogeneity of soils, plant competition within plots, climatic factors, genetic differences, etc. Such factors are termed as extraneous factors.

There are different types of experimental designs like

1. Single factors experiments
2. Multi-factor (factorial) experiments

The single factor experiments can be grouped as complete block designs and incomplete block designs.

When the treatment consist different levels of a single variable factor and all other factors are kept at a single prescribed level et is known as a single factor experiment.

Example

In fertilizer trials several rates of single fertilizer element say nitrogen may be tested. All other factors such as agronomic practices, water management, insect control etc are kept at a uniform level.

Basic principles of Experimental Design

Basic Principles

The basic principles of any design of experiment are replication, randomization and local control of error.

Replication in an Experiment: The repetition of the experiment in different experimental unit is known as replication. If the treatment is applied only once we have no means of knowing about the variations in the results of treatment. Only when we repeat the application of treatment several times we can estimate the experimental error. Only with the help of experimental error, we can determine whether observed differences between treatment means are real or not. The number of replications are increased the experimental error will be reduced.

The advantages of replication are:

1. Replications are essential to obtain a valid estimate of the experimental error variance.
2. To reduce the experimental error. The larger number of replications reduces the standard error of the treatment means but one cannot take more than certain number of replications because of the management problems and limitation of resources
3. It eliminates the systematic bias
4. It is needed to obtain a representative sample from the population

Randomization: When all the treatments have an equal chance of being allocated to different experimental unit it is known as randomization. The procedure of allocating treatment to experimental unit may vary according to the design of the experiment. The advantages of randomization are:

1. To get an unbiased estimate of experimental error
2. To get valid conclusions

Local control (Error control)

Local control is a device to maintain greater homogeneity of experimental units a block of an experiment (or) as a whole. For instance, Soil fertility of field is a factor that affects the plant growth and yield. So all the neighbor plots have the same soil fertility should constitute a block. Conducting a uniformity trial can assess the soil fertility of land. Before conducting the actual field experiment we prepare the fertility contour map on the basis of fertility gradient. These contours help to form blocks. Local control is also known as error control. It increases the efficiency of an experiment. The advantages of local control are:

1. Local control reduces the experimental error
2. Local control is meant to make designs more efficient
3. It makes any test of significance more sensitive and powerful
4. A reduction in Experimental error consequently helps the investigator to detect the small real difference between treatment means.

Uniformity Trial

To achieve local control knowledge of the experimental material is a must. This is conducted to know the soil fertility gradient. The uniformity trial data may also be used to determine the size of the experimental unit. In uniformity trial a particular variety of a crop is sown on the entire experimental field and uniformly managed throughout the growing season. The field is divided into small plots termed as basic units after leaving out a substantial border. The data is recorded on each basic unit. Then the mean yield per basic unit is calculated. The basic units giving yields above or below a specified percentage of the overall mean, say 5% or 10% are marked on the plan of the plots. The similar units are then joined by lines to produce the contour map. Such a map is called fertility contour map or fertility gradient map or soil fertility

map. Using this map, homogenous experimental units can be grouped as blocks. To determine the optimum plot size and shape maximum curvature method and Fairfield Smith's variance law can be used.

ANOVA (Analysis of Variance)

The ANOVA is a powerful statistical tool for tests of significance. The test of significance based on t-distribution is an adequate procedure only for testing the significance of the difference between two sample means. In a situation when we have two or more samples to consider at a time, an alternative procedure is needed for testing the hypothesis that all the samples have been drawn from the same population. For example, if three fertilizers are to be compared to find their efficacy, this could be done by a field experiment, in which each fertilizer is applied to 10 plots and then the 30 plots are later harvested with the crop yield being calculated for each plot. Now we have 3 groups of ten figures and we wish to know if there are any differences between these groups. The answer to this problem is provided by the technique of ANOVA. The term ANOVA was introduced by Prof. R.A. Fisher in 1920ˇs to deal with problem in the analysis of agronomical data. Variation is inherent in nature

Definition: The anova is a simple arithmetic process of sorting out the components of variation in a given data

Assumptions

- The observations are independent
- The experimental error are independent and normally distributed with mean zero and variance 1
- Parent population from which observations are taken is normal
- Various treatment and environment effect are additive in nature

Uses

- To test the homogeneity of several means
- To compare the means of three or more samples

Types of anova

There are two types

(1) One way classification (2) two way classification

Types of Experimental Designs

- Completely Randomized Design
- Randomized Block Design
- Factorial Experiments

Completely Randomized Design (Single factor Experiment)

The Completely Randomized (CRD) design randomly divides the experimental units into t groups of size n and randomly assigns a treatment to each group. When the treatments are

arranged randomly over the whole of a previously determined set of experimental units the design in known as completely randomized design. CRD is appropriate only when the experimental material is homogeneous. In field experiments there is generally large variation among experimental plots due to soil heterogeneity. Hence CRD is not preferred in field experiments. In laboratory experiments and greenhouse studies it is easy to achieve homogeneity of experimental materials. Therefore CRD is most useful in such experiments
Ex: in fertilizer trials several rates of single fertilizer element say nitrogen, may be tested

Advantages:

- The design is most commonly used in laboratory experiment such as Ag. Chemistry, Plant pathology and animal experiments where the experimental material is expected to be homogenous
- The design is useful in pot cultural experiments where the same type of soil is usually used
- Any number of replications and treatments can be used. The number of replication is vary from treatment to treatment
- The analysis remain simple even is information of some units are missing
- The design provides maximum number of degrees of freedom for the estimation of error then the other designs

Disadvantages

It is difficult to find homogeneous experimental units in all aspects
CRD is seldom suitable for field experiments as compared to other experimental designs

Statistical Model for CRD

The statistical model for CRD with one observation per unit

$Y_{ij} = \mu + t_i + e_{ij}$

μ = overall mean effect

t_i = true effect of the ith treatment

e_{ij} = error term of the jth unit receiving i^{th} treatment

Procedural Overview

H_0: There is no significant difference between the columns
H_1: There is significant difference between the columns
Step 1: Find N (Number of observations) and T (Total number of all observations)

Step 2: Correction factor = $\frac{T^2}{N}$

Step 3: TSS= $\sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 + ... - \frac{T^2}{N}$

Step 4: SSC= $= \frac{\left(\sum x_1\right)^2}{N_1} + \frac{\left(\sum x_2\right)^2}{N_1} + \frac{\left(\sum x_3\right)^2}{N_1} + \frac{\left(\sum x_4\right)^2}{N_1} + \ldots - \frac{T^2}{N}$

Here N_1 is the number of elements in each column

SSE=TSS-SSC

Step 5: ANOVA table

Source of Variation	Sum of squares	Degree of Freedom	Mean sum of squares	Variance ratio	Table value at 5% level
Between Columns	SSC	C-1	$MSC = \frac{SST}{C-1}$	$F_{Cal} = \frac{MSC}{MSE}$	$F_{tab} = F_C(C-1, N-C)$
Error or Residual	SSE	N-C	$MSE = \frac{SSE}{N-C}$		

Conclusion: Since $F_{Cal} > F_{Tab}$ we reject H_0 and conclude that there is a significant difference between the treatment means otherwise it is accepted

Layout for CRD

The arrangement of data in CRD is as follows:

	Treatments				
	T_1	T_2	T_i	T_K	
	y_{11}	y_{21}	y_{i1}	Y_{K1}	
	y_{12}	y_{22}	y_{i2}	Y_{K2}	
	y_{1r1}	y_{2r2}	y_{iri}	$Y_{k\,rk}$	
Total	Y_1	Y_2	Y_i	T_k	GT

(GT – Grand total)

Randomized Block Design

The experimental material is divided into r blocks. Let there are t treatments. Each block is then divided into t units and the treatment is allocated within the block. In agricultural field experiments the experimental units are not homogenous and hence the design CRD cannot be adopted. So when the experimental material is heterogeneous, the experimental material is grouped into homogenous sub-groups called blocks. Since each block consists of the entire set of treatments a block is equivalent to a replication. If the fertility gradient runs in one direction say from north to south or east to west then the blocks are formed in the opposite direction. Such an arrangement of grouping the heterogeneous units into homogenous blocks is known as randomized blocks design. Sub group is commonly termed as block. Each block consists of entire set of treatments. A block is equivalent to replication.

The treatments are allocated randomly to the experimental units within each block independently such that each treatment occurs once. So each treatment is replicated as many times as the number of blocks or the number of blocks are chosen to be equal to the number of replications for the treatments.

Example

Slope of the field, in a study of plant reaction to water stress

Advantages of RBD

The precision is more in RBD. The amount of information obtained in RBD is more as compared to CRD. RBD is more flexible. Statistical analysis is simple and easy. Even if some values are missing, still the analysis can be done by using missing plot technique

Disadvantages of RBD

- Unequal number of replications for equal treatment is not possible
- It controls the variability in one direction only
- The optimal blocks size in field experiment is 21 plots
- The shape of the experimental material should be square or rectangular

Treatment range 2 to 24

Statistical Model for RBD

The analysis of variance model for RBD is

$Yij = \mu + ti + rj + eij$

where

μ = the overall mean

t_i = the ith treatment effect

r_j = the jth replication effect

eij = the error term for i^{th} treatment and j^{th} replication

Layout for RBD

Treatment	Replication					Total
	1	2	3	.....	r	
1	y11	y12	y13	.....	y1r	T1
2	y21	y22	y23	.....	y2r	T2
3	y31	y32	y33	.....	y3r	T3
t	yt1	yt2	yt3	.....	ytr	Tt
Total	R1	R2	R3		Rr	G.T

Working procedure

H_0: There is no significant difference between the Rows and columns effect

H_1: There is a significant difference Rows and columns effect

Step 1: Find N (Number of observations) and T (Total number of all observations)

Step 2: Correction factor = $\frac{T^2}{N}$

Step 3: TSS= $\sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 + ... - \frac{T^2}{N}$

Step 4: SSC= $= \frac{\left(\sum x_1\right)^2}{N_1} + \frac{\left(\sum x_2\right)^2}{N_1} + \frac{\left(\sum x_3\right)^2}{N_1} + \frac{\left(\sum x_4\right)^2}{N_1} + \ldots - \frac{T^2}{N}$

Step 5: SSR = $= \frac{(\sum y_1)^2}{N_2} + \frac{(\sum y_2)^2}{N_2} + \frac{(\sum y_3)^2}{N_2} + \frac{(\sum y_4)^2}{N_2} + \ldots - \frac{T^2}{N}$

Here N_2 is the number of elements in each Row

Step 6: SSE=TSS-SSC-SSR

Step 7: ANOVA table

Source of Variation	Sum of squares	Degree of Freedom	Mean sum of squares	Variance ratio	Table value at 5% level
Between Columns	SSC	c-1	$MSC = \frac{SST}{c-1}$	$F_c = \frac{MSC}{MSE}$	$F_{tab} = F_C(c-1,(r-1)(c-1))$
Between Rows	SSR	r-1	$MSR = \frac{SSR}{r-1}$	$F_r = \frac{MSR}{MSE}$	$F_{tab} = F_r(r-1,(r-1)(c-1))$
Error or Residual	SSE	(r-1)(c-1)	$MSE = \frac{SSE}{(r-1)(c-1))}$		

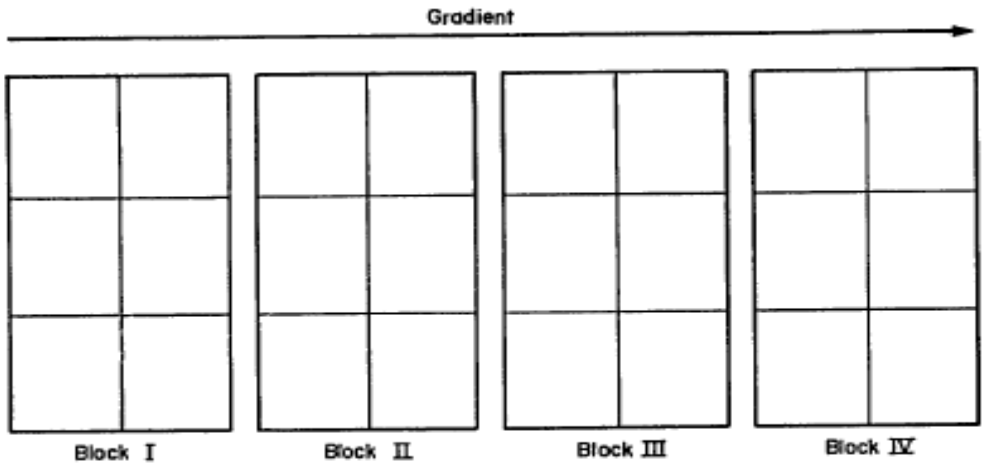


Figure 2.2 Division of an experimental area into four blocks, each consisting of six plots, for a randomized complete block design with six treatments and four replications. Blocking is done such that blocks are rectangular and perpendicular to the direction of the unidirectional gradient (indicated by the arrow).

Latin Square Design

The major feature of the LS design is its capacity to simultaneously handle two known sources of variation among experimental units. It treats the sources as two independent blocking criteria, instead of only one as in the RBD. The two directional blocking in a LS design, commonly referred to as row blocking and column blocking is accomplished by ensuring that every treatment occurs only once in each row block and once in each column clock.

When the experimental material is divided into rows and columns and the treatments are allocated such that each treatment occurs only once in each row and each column, the design is known as L S D

Advantages

- LSD controls variability in two directions of the experimental material
- The analysis of the design is simple and straight forward
- When missing values are present, missing plot technique can be used and analysed.
- LSD is more efficient than RBD or CRD. This is because of double grouping that will result in small experimental error

Disadvantages

- The process of randomization is not as simple as in RBD
- LSD is not suitable for treatments less than five.
- LSD is suitable only in the case of smaller number of treatments
- A 2x2 latin square is not possible

Layout

Row Number	Column Number				
	1	2	3	4	5
1	*E*	*C*	*B*	*A*	*D*
2	*A*	*D*	*C*	*B*	*E*
3	*C*	*B*	*D*	*E*	*A*
4	*B*	*E*	*A*	*D*	*C*
5	*D*	*A*	*E*	*C*	*B*

Statistical Model for LSD

$$Y_{ijk} = \mu + r_i + c_j + t_k + e_{ijk}$$

r_i is the ith row effect

c_j is the jth column effect

t_k is the kth treatment effect and

eijk is the error term

Working procedure

H_0**:** There is no significant difference between the Rows, Columns and treatment effects
H_1: There is a significant difference Rows Columns and treatment effects

Step 1: Find N (Number of observations) and T (Total number of all observations)

Step 2: Correction factor = $\frac{T^2}{N}$

Step 3: TSS= $\sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 + ... - \frac{T^2}{N}$

Step 4: SSC= $= \frac{(\sum x_1)^2}{N_1} + \frac{(\sum x_2)^2}{N_1} + \frac{(\sum x_3)^2}{N_1} + \frac{(\sum x_4)^2}{N_1} + \ldots - \frac{T^2}{N}$

Step 5 : SSR = $= \frac{(\sum y_1)^2}{N_2} + \frac{(\sum y_2)^2}{N_2} + \frac{(\sum y_3)^2}{N_2} + \frac{(\sum y_4)^2}{N_2} + \ldots - \frac{T^2}{N}$

Here N_2 is the number of elements in each Row

Step 6: After arranging the observations and make the total row wise

SST= $\frac{(T_1)^2}{N_2} + \frac{(T_2)^2}{N_2} + \frac{(T_3)^2}{N_2} + \ldots - \frac{T^2}{N}$

Step 7: SSE=TSS-SSC-SSR-SST

Step 8: ANOVA Table

Source of Variation	Sum of squares	Degree of Freedom	Mean sum of squares	Variance ratio	Table value at 5% level
Between Columns	SSC	t-1	$MSC = \frac{SSC}{t-1}$	$F_C = \frac{MSC}{MSE}$	$F_{tab} = F_C(t-1,(t-1)(t-2))$
Between Rows	SSR	t-1	$MSR = \frac{SSR}{t-1}$	$F_R = \frac{MSR}{MSE}$	$F_{tab} = F_R(t-1,(t-1)(t-2))$
Between treatments	SST	t-1	$MST = \frac{SST}{t-1}$	$F_T = \frac{MST}{MSE}$	$F_{tab} = F_T(t-1,(t-1)(t-2))$
Error or Residual	SSE	(t-1)(t-2)	$MSE = \frac{SSE}{(t-1)(t-2)}$		

Conclusion: Since $F_{Cal} > F_{Tab}$ we reject H_0 and conclude that there is a significant difference between the treatment means otherwise it is accepted

STA 201 THEORY TYPE QUESTION

1. Explain various measures of central tendency
2. Explain Various measure of dispersion
3. Explain the properties of Binomial, Poisson and Normal distribution
4. Explain probability and non-probability sampling
5. Write the procedure for testing of hypothesis
6. Explain basic principles of Experimental design
7. Explain the regression analysis in detail
8. Define hypothesis and write the detailed procedure for testing the association between attributes(i.e Chi-square test)
   OR Which test is applied in case of testing independence of attributes and write stepwise procedure to perform this test
9. Describe correlation analysis
10. Explain CRD in details
11. Explain RBD in details
12. Write down the procedure for test the difference between two sample means
13. Write detailed note on chi square test
14. Give the procedure for testing the one and two sample proportions
15. Explain the test procedure for testing equality two means in small samples
16. Describe a LSD
17. Write the steps involved in testing of hypothesis with an examples of t-test for single mean

A

TAMIL NADU AGRICULTURAL UNIVERSITY
B.Sc. (Hons.) Agri. / Horti. / Seri. / Forestry / FND / B.Tech. (AIT / Bioinfo. / Biotech.)
B.S. (ABM) Degree Programme
Final Theory Examination, August 2019

Year : II **(2017 Syllabus)** **Time : 2 ½ hr**
Semester : IV **Max. Marks: 40**

STA 201 Statistical Methods (1+1)
PART-A **[20 x 0.5 = 10]**
(Answer **All** questions)

Choose the correct answer

1. The word 'statistics' is used as
   A) Singular alone B) Plural alone
   C) Singular and Plural both D) Either singular or plural

2. A sample consists of
   A) All units of population B) 50% units of the population
   C) 5% units of the population D) Any fraction of the population

3. Power of test is related to
   A) Type I error B) Type II error
   C) Type I and Type II error D) Degree of freedom

4. The term regression was introduced by
   A) R. A. Fisher B) Sir Francis Galton
   C) Karl Pearson D) Mahalanobis

5. Replication in an experiment means
   A) The number of blocks B) Total number of treatments
   C) The number of times a treatment occurs in an experiment D) The no. of missing values

Fill in the blanks with correct answer

6. Geometric mean is suitable when the values are given as ----------
   A) Ratio B) Proportion
   C) Both A and B D) Either A or B

7. The number of parameters involved in normal distribution p.d.f is ----------
   A) 2 B) 3
   C) 4 D) 1

8. Probability of Type I error is called ----------
   A) Level of significance B) Power of the test
   C) Both A and B D) Degree of freedom

9. If the correlation coefficient is zero, the value of regression coefficient is ------
   A) Zero B) One
   C) Infinite D) Any value between zero to infinite

10. The allocation of treatments to experimental units with equal probability is known as----------
    A) Randomization B) Replication
    C) Local Control D) Uniformity trail

Match the following

11. Range of Coefficient of variation
    A) 0 to 100
    B) 0 to ∞
    C) 0 to 1
    D) -∞ to +∞

12. A and B are two independent events P(AUB)
    A) P(A)P(B)
    B) P(A)+P(B)-P(A∩B)
    C) P(A)+P(B)
    D) P(B)

13. Test statistic for two variance
    A) F test
    B) t test
    C) Z test
    D) X^2 test

14. Range of Regression coefficient
    A) 0 to 100
    B) 0 to ∞
    C) 0 to 1
    D) -∞ to +∞

15. Error degrees of freedom in RBD with r replication and t treatments
    A) rt-1
    B) rt-t
    C) rt-r
    D) (r-1)(t-1)

State Correct/ Incorrect Statement

16. Identify the **correct** statement
    A) If 5 is subtracted from each observation of a set, then the mean of the observation is increased by 5
    B) If 5 is subtracted from each observation of a set, then the mean of the observation is decreased by 5
    C) If 5 is subtracted from each observation of a set, then the mean of the observation is increased by square root of 5
    D) None of the above

A

17. Identify the **correct** statement
A) Probability of sample space is equal to one
B) Probability of sample space is equal to zero
C) Probability of sample space is equal to 100
D) Probability of sample space is equal to infinite

18. Identify the **Incorrect** statement
A) Rejecting H_0 when H_0 is true is type I error
B) Rejecting H_0 when H_0 is false is type I error
C) accepting H_0 when H_0 is false is type I error
D) accepting H_0 when H_0 is true is type II error

19. Identify the **correct** statement
A) Correlation coefficient is always positive
B) Correlation coefficient is always negative
C) Correlation coefficient lies between $-\infty$ to $+\infty$
D) Correlation coefficient lies between -1 to +1

20. Identify the **correct** statement
A) CRD can be used in green house condition
B) CRD can be used in field condition
C) RBD can be used in green house condition
D) RBD can be used in laboratory condition

PART – B
(Answer any **Five** questions) **[5 x 1 = 5]**

21. Which of these is not a two dimensional diagram
A) Rectangles B) Cylinder
C) Circles D) Pie diagram

22. The limiting relative frequency approach of probability is known as
A) Statistical probability B) Classical probability
C) Mathematical probability D) All the above

23. A sample of 12 specimens taken from a normal population is expected to have a mean 50 mg/cc. The sample has a mean 64 mg/cc with a variance of 25. To test H_0: $\mu=50$ vs. $H_1:\neq 50$, you will use
A) Z test B) X^2
C) F test D) t test

24. If correlation coefficient is equal to zero, the angle between the two lines of regression is
A) Zero degree B) Ninety degree
C) Sixty degree D) Thirty degree

25. Randomized block design is a
A) Three restrictional design B) Two restrictional design
C) One restrictional design D) No restrictional design

A

26. Missing observation in a completely randomized design is to be
A) Estimated B) Deleted
C) Guessed D) None of the above

PART – C
(Answer any **Five** questions) **[5 x 2 = 10]**

C1. What do you mean by classification? Mention its kinds.

C2. If X follows binomial distribution (X~B(10,1/2)), then mean and variance of the variable X is

C3. In which situation sampling is inevitable?

C4. Write note on two related sample test.

C5. Distinguish between correlation and regression?

C6. Fill the blank

SV	Df	ss	ms	Fcal
Replication	4	40	10	5
treatment	7	84	12	6
Error	28	56	2	-
Total	39	180	-	-

PART – D
(Answer any **Three** questions) **[3 x 5 = 15]**

D1. X = 20, 25, 24, 30, 28, 22, 29,26, 23, 27 then calculate Mean, variance, standard deviation, range and coefficient of variation for X.

D2. Describe simple random sampling in detail?

D3. What is correlation? What do you understand by scattered diagram? Explain in detail about Karl Pearson correlation coefficient and its properties.

D4. An experiment was conducted using six treatments and four replication, yield of the crop was recorded

TREATMENT	R1	R2	R3	R4
T1	19	38	32	42
T2	29	46	40	41
T3	33	36	51	51
T4	44	45	56	54
T5	45	50	55	56
T6	54	50	52	54

Analyze the data and prepare ANOVA table.

****

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